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A Third-Order Structure-Preserving Exponential Time Differencing Runge-Kutta Scheme for the Binary Fluid-Surfactant Phase Field Model

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Abstract
In this paper, we present and analyze a third-order (in time) numerical scheme for a binary fluid-surfactant phase field model which consists of two coupled Cahn-Hilliard type equations. The proposed scheme inherits the structure-preserving property of the model, including energy dissipation and bound preservation. To maintain the dissipation with respect to the original free energy, a linear convex splitting of the energy functional is first introduced. The numerical scheme is then developed by employing a spectral collocation approximation for spatial discretization and a third-order exponential time differencing Runge-Kutta scheme for time integration. Benefiting from the explicit evaluations of the nonlinear and coupling terms, the proposed scheme is linear and completely decoupled. By treating the numerical solution as the sum of the exact solution and a small perturbation, the bound-preserving property of the proposed scheme is proved along with its convergence. The original energy dissipation is further achieved with the help of the convex splitting method and the control of the nonlinear and coupling terms. Some numerical experiments in two and three dimensions are performed to verify the theoretical results and demonstrate the efficiency of the proposed scheme for simulations of phase separation phenomena.

Keywords Binary fluid-surfactant model · Linear convex splitting · Third-order exponential time differencing · Structure-preserving

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1 Introduction

Surfactants, with the amphiphilic nature derived by their hydrophilic and hydrophobic groups, play a crucial role in industries such as detergent manufacturing, pharmaceuticals, and oil exploitation. In this paper, we consider the phase field model for the binary fluid-surfactant (BFS) system. The model reads as a gradient flow driven by the energy functional developed in [7] without hydrodynamical effects:

$$G(\phi,\rho)=\int_{\Omega}\left(\frac{\eta_1}{2}|\Delta\phi|^2+\frac{\varepsilon}{2}|\nabla\phi|^2+\frac{f(\phi)}{\varepsilon}+\frac{\eta_2}{2}|\nabla\rho|^2+h(\rho)-\theta\rho|\nabla\phi|^2\right)\mathrm{d}\boldsymbol{x}. \quad (1)$$

Here $\Omega\subseteq\mathbb{R}^d$ $(d=2,3)$, η_1, η_2, ε and θ are all positive parameters, and

$$f(\phi)=\frac{1}{4}(\phi^2-1)^2,\quad h(\rho)=\beta[\rho\ln\rho+(1-\rho)\ln(1-\rho)]+\frac{\beta_1}{2}\rho(1-\rho),$$

in which β is a positive parameter, and $\beta_1\in\mathbb{R}$. The functions $\phi:\Omega\to\mathbb{R}$ and $\rho:\Omega\to\mathbb{R}$ are two phase field variables representing the fluid density and surfactant concentration, respectively, with $\phi=-1$ and $\phi=1$ indicating different fluid phases. The Ginzburg-Landau double well potential $f(\phi)/\varepsilon$ promotes the formation of two distinct phases within the system, and the gradient term $|\nabla\phi|^2$ introduced in [24] minimizes the area of the interfaces. Both terms contribute as the driving force of the coarsening dynamics. The Flory-Huggins potential $h(\rho)$ includes the mixing entropy and the lateral interaction of surfactants in fluids. Moreover, the term $|\nabla\rho|^2$ can ensure that the resulting system is well-posed. For more details on the energy terms of the surfactant concentration, see [12] and the references therein. The nonlinear coupling entropy $-\theta\rho|\nabla\phi|^2$ has been widely used to describe the adsorption behavior of surfactants near the interfaces [23, 30, 33], while some other forms of the coupling terms have also been adopted in [12, 14, 32, 39]. Moreover, since the surfactant concentration near the interfaces can be higher than that in the pure phase domains, the inclusion of the high-order diffuse term $|\Delta\phi|^2$ is necessary to prevent the energy from being unbounded from below [24], which ensures the well posedness of the model.

It is still a challenging task to construct efficient and structure-preserving (energy-dissipative and bound-preserving) numerical schemes for time stepping of the BFS phase field model due to the presence of complicated nonlinear terms in the energy functional, particularly the nonlinear coupling entropy. A commonly used method to develop energy-dissipative (or called energy-stable) numerical schemes for phase field models is the convex splitting method first proposed by Eyre in [13]. The idea is to split the energy into convex and concave parts and treat these two parts implicitly and explicitly, respectively. In [17], a first-order numerical scheme based on the convex splitting method was constructed for the BFS system, but its convex-concave splitting of the energy was defective, which fails to show the convexity of the multivariate energy density. Another first-order scheme based on this method was developed in [28], where the positivity preservation $(0<\rho<1)$ and optimal order convergence were also provided. Although the convex splitting method ensures the energy dissipation of the constructed schemes, the implicit treatment of the nonlinear terms results in nonlinear schemes, which need to be solved iteratively and incur high computational costs. To develop linear numerical schemes, the invariant energy quadratization (IEQ) method was adopted in [40, 41] to construct first- and second-order schemes for the BFS system. As a variant of the IEQ method, the scalar auxiliary variable (SAV) method

has also been applied in recent years to develop decoupled linear schemes up to the second order for this system [37, 39, 43]. However, due to the introduction of auxiliary variables, the energy dissipation obtained by these two numerical methods is with respect to certain alternate energies modified from the original one. The convergence and bound preservation of the numerical solutions were also not analyzed in these studies. More recently, to design both linear and original energy-dissipative schemes, the exponential time differencing (ETD) method was applied to construct first- and second-order numerical schemes for the BFS system [8, 11], whose energy dissipation was proved following the similar spirit from [9, 16]. Additionally, the convergence analysis was conducted in [8, 11], and the analysis for the second-order ETD schemes for other phase field models can be found in [9, 25, 26, 31].

There also exist some high-order (greater than two) numerical schemes for the BFS model. In [18], a first-order scheme for the BFS system was constructed by a combination of the discontinuous Galerkin method, convex splitting method and IEQ method. By employing the spectral deferred correction method iteratively, the third-order temporal accuracy was observed in numerical tests, but no theoretical proof was provided. In addition, based on the operator splitting approach, the extrapolation method was applied for the time integration to develop a fourth-order scheme for the BFS system in [36]. Nevertheless, the analysis for both structure preservation and convergence were again not conducted.

In this paper, we will apply the ETD approach to construct a linear third-order and structure-preserving scheme for the BFS model. The ETD method exhibits the ability to exactly evaluate the linear part of systems and provides remarkable stability, even in the presence of strong stiffness exhibited by the linear operator [1, 6, 20, 21]. Moreover, the ETD schemes can be implemented by fast Fourier transform (FFT)-based fast algorithms for computing operator exponentials in rectangular domains. Due to these advantages, the ETD method has been employed in a diverse array of phase field models [9, 10, 22, 26, 27, 42]. In fact, some high-order ETD multistep schemes have been developed and analyzed in [3–5] for the thin film growth equation or gradient flows with Lipschitz continuous nonlinear terms; however, the energy dissipation of these schemes is all derived in terms of modified energies. Recently, a general framework was conducted in [15] to establish the original energy dissipation of high-order ETD Runge-Kutta (ETDRK) schemes for a single phase field equation. The authors have discovered that some three-stage third-order ETDRK schemes listed in [20, 21], when equipped with appropriate coefficients, could meet the requirements of the framework. A new third-order ETDRK scheme has also been proposed in [2] to achieve the original energy dissipation. Furthermore, in [38], the authors proposed new parametric ETDRK schemes which regard the third-order scheme designed in [2] as a special case, and introduced a novel approach based on semi-positive definite matrix decomposition to establish the original energy stability when the parameter is within a certain range.

The main contribution of this paper is the development and analysis of a third-order ETDRK scheme for the BFS model, which inherits the structure-preserving properties of the system. By treating the numerical solution as a combination of the exact solution and a small perturbation, the convergence and boundedness of the numerical solutions are derived inductively. Combining the linear convex splitting approach with the framework developed in [15], we further utilize the boundedness of the numerical solutions rather than assuming the Lipschitz continuity of the nonlinear term as in [15] to show the original energy dissipation of the proposed scheme. To achieve high-order accuracy in space, the spectral collocation method is employed to spatial discretization. The proposed scheme is linear and completely decoupled due to the explicit treatment of nonlinear terms in the ETD method. By considering the periodic boundary condition, the scheme can be implemented using the FFT in a rectangular domain, resulting in highly efficient computations.

The rest of the paper is organized as follows. In Section 2, we briefly introduce the system of BFS phase field model. The fully-discrete ETDRK scheme is constructed in Section 3. In Section 4, we show the convergence and bound-preserving property of the proposed scheme, and its energy dissipation property is then proved in Section 5. In Section 6, some numerical experiments in two and three dimensions are performed to verify the theoretical convergence and demonstrate the efficiency of the proposed scheme. Finally, concluding remarks are given in Section 7.

2 The Phase Field Model for the Binary Fluid-Surfactant System

The system of BFS phase field equations corresponds to the H^{-1} gradient flow of the free energy functional (1) and can be formulated as follows:

$$\phi_t = \Delta\Big(\eta_1 \Delta^2 \phi - \varepsilon \Delta \phi + \frac{1}{\varepsilon} f'(\phi) + 2\theta \nabla \cdot (\rho \nabla \phi)\Big), \tag{2a}$$

$$\rho_t = \Delta\big(-\eta_2 \Delta \rho + h'(\rho) - \theta |\nabla \phi|^2\big), \tag{2b}$$

where $\phi = \phi(\boldsymbol{x}, t) : \overline{\Omega} \times [0, \infty) \to \mathbb{R}$ and $\rho = \rho(\boldsymbol{x}, t) : \overline{\Omega} \times [0, \infty) \to \mathbb{R}$ are unknown functions. In this work, we assume that ϕ and ρ are periodic for simplicity. In [7], the system (2) equipped with hydrodynamical effects is proved to be well-posed, in which the homogeneous Neumann boundary conditions are applied for ϕ and ρ. The well-posedness of the system (2) equipped with the periodic boundary conditions can be derived similarly. The system (2), which contains two coupled Cahn-Hilliard type equations, satisfies the energy dissipation law in the sense that

$$\frac{\mathrm{d}}{\mathrm{d}t} G(\phi(t), \rho(t)) \leq 0.$$

A uniform bound for the phase variable ρ, i.e., $0 < \rho < 1$ is implied by the logarithmic terms of the potential $h(\rho)$. We aim to develop a third-order (in time) numerical scheme that is structure-preserving, i.e., its numerical solutions not only inherit the energy dissipation but also uniformly preserve the bounds of ρ and ϕ.

3 Fully-Discrete Exponential Time Differencing Runge-Kutta Scheme

In this section, we present the third-order (in time) fully-discrete numerical scheme for the BFS model (2). We first conduct a linear convex splitting of the energy functional (1), then we discretize the resulting system in space and time using the spectral collocation method and the ETDRK method, respectively.

3.1 Linear Convex Splitting

By introducing three stabilization terms, we propose the following linear convex splitting of the energy functional (1):

$$G(\phi, \rho) = G_c(\phi, \rho) - G_e(\phi, \rho)$$

with

$$G_c(\phi,\rho)=\int_\Omega\left(\frac{\eta_1}{2}|\Delta\phi|^2+\frac{\varepsilon}{2}|\nabla\phi|^2+\frac{A_1}{2\varepsilon}\phi^2+\frac{\eta_2}{2}|\nabla\rho|^2+\frac{B}{2}\rho^2+\frac{A_2}{2}|\nabla\phi|^2\right)\mathrm{d}\boldsymbol{x},$$
$$G_e(\phi,\rho)=\int_\Omega\left(-\frac{f(\phi)}{\varepsilon}+\frac{A_1}{2\varepsilon}\phi^2-h(\rho)+\frac{B}{2}\rho^2+\theta\rho|\nabla\phi|^2+\frac{A_2}{2}|\nabla\phi|^2\right)\mathrm{d}\boldsymbol{x}$$

where A_1, A_2 and B are positive parameters. It is easy to see that $G_c(\phi,\rho)$ is convex. For the convexity of $G_e(\phi,\rho)$, we have the following result:

Proposition 1 *Suppose that there exist positive constants $K_\phi\geq 1$, $\widetilde{K}_\phi>0$ and $\frac{1}{2}<K_\rho<1$ such that $|\phi|\leq K_\phi$, $|\nabla\phi|\leq\widetilde{K}_\phi$ and $1-K_\rho\leq\rho\leq K_\rho$ in Ω. If*

$$A_1\geq 3K_\phi^2-1,\quad A_2\geq 2\theta\widetilde{K}_\phi,\quad B\geq 2\sqrt{2}\theta\widetilde{K}_\phi+\frac{\beta}{K_\rho(1-K_\rho)}-\beta_1,\tag{3}$$

then the energy $G_e(\phi,\rho)$ is convex with respect to ϕ and ρ.

Proof Let $\boldsymbol{v}=(v_1,v_2,v_3,v_4)=(\phi,\rho,\phi_x,\phi_y)$ and define

$$e_e(\boldsymbol{v})=-\frac{f(\phi)}{\varepsilon}+\frac{A_1}{2\varepsilon}\phi^2-h(\rho)+\frac{B}{2}\rho^2+\theta\rho|\nabla\phi|^2+\frac{A_2}{2}|\nabla\phi|^2,$$
$$e_{e1}(\boldsymbol{v})=-\frac{f(\phi)}{\varepsilon}+\frac{A_1}{2\varepsilon}\phi^2,$$
$$e_{e2}(\boldsymbol{v})=-h(\rho)+\frac{B}{2}\rho^2+\theta\rho|\nabla\phi|^2+\frac{A_2}{2}|\nabla\phi|^2.$$

By the condition (3), we have

$$\partial_{v_1}^2 e_{e1}(\boldsymbol{v})=\frac{1}{\varepsilon}\big(A_1-(3\phi^2-1)\big)\geq 0,$$

and thus $e_{e1}(\boldsymbol{v})$ is convex. The Hessian matrix of $e_{e2}(\boldsymbol{v})=e_{e2}(v_2,v_3,v_4)$ can be written as

$$H=\begin{bmatrix}B+\beta_1-\frac{\beta}{v_2(1-v_2)} & 2\theta v_3 & 2\theta v_4\\ 2\theta v_3 & 2\theta v_2+A_2 & 0\\ 2\theta v_4 & 0 & 2\theta v_2+A_2\end{bmatrix}$$

H is diagonally dominated with all diagonal entries nonnegative when the conditions in (3) are met, which implies the convexity of $e_{e2}(\boldsymbol{v})$. Since $e_e(\boldsymbol{v})=e_{e1}(\boldsymbol{v})+e_{e2}(\boldsymbol{v})$, we know that $e_e(\boldsymbol{v})$ is also convex so that the convexity of $G_e(\phi,\rho)$ is obtained. □

Remark 1 The linear convex splitting method has already been adopted to construct energy stable ETD schemes in [8, 11]. In [8], the convexity of the energy $G_e(\phi,\rho)$ was proved by modifying the potentials to make their second derivatives bounded on the whole real line. However, the convexity of $G_e(\phi,\rho)$ is claimed in terms of each variable with others fixed, which is not sufficient to ensure the convexity of the multivariate energy. In [11], the convexity of the energy was ensured by choosing appropriate stabilization parameters such that the Hessian matrix of the energy density $e_{e2}(\boldsymbol{v})$ is diagonally dominant, to which our proof is similar. However, the choice of stabilization parameters in [11] relies on the boundedness of ϕ and ρ and a small introduced δ, rather than the boundedness of ϕ, ρ and $|\nabla\phi|$ stated in Proposition 1 in our work. Since a different nonlinear coupling entropy was used in [11], the δ was added to the entropy to avoid the singularity arising from the denominator of the resulting coupling term in the evolution system, which slightly changed the whole system.

Although choosing a smaller δ can make the system deviate less from the original system, the convergence tests in [11] demonstrated that it can lead to larger numerical errors. To avoid these two drawbacks, we adopt the BFS model equipped with the coupling entropy $-\theta\rho|\nabla\phi|^2$, which does not result in the singularity as noted in [11].

3.2 Spectral Collocation Approximations for Spatial Discretization

Let us consider $\Omega = (0, L_x) \times (0, L_y)$. Let N_x and N_y be two positive even numbers and define the mesh sizes as $h_x = L_x/N_x$ and $h_y = L_y/N_y$ in each dimension. Denote by $\Omega_{\mathcal{N}}$ the $N_x \times N_y$ uniform mesh of the domain Ω as

$$\Omega_{\mathcal{N}} = \{(x_i, y_j) = (ih_x, jh_y) \mid 1 \le i \le N_x,\ 1 \le j \le N_y\},$$

and we define the following space of periodic functions:

$$\mathcal{M}_{\mathcal{N}} = \{f : \mathbb{Z}^2 \to \mathbb{R} \mid f_{i+pN_x, j+qN_y} = f_{ij},\ \forall\,(p, q) \in \mathbb{Z}^2\}.$$

Denote the index sets

$$S_{\mathcal{N}} = \{(i, j) \in \mathbb{Z}^2 \mid 1 \le i \le N_x,\ 1 \le j \le N_y\},$$
$$\widehat{S}_{\mathcal{N}} = \{(k, l) \in \mathbb{Z}^2 \mid -\frac{N_x}{2} + 1 \le k \le \frac{N_x}{2},\ -\frac{N_y}{2} + 1 \le l \le \frac{N_y}{2}\}.$$

For a function $f \in \mathcal{M}_{\mathcal{N}}$, the two-dimensional discrete Fourier transform $\widehat{f} = Pf$ is defined componentwise [29, 34] by

$$\widehat{f}_{kl} = \frac{1}{N_x N_y} \sum_{(i,j)\in S_{\mathcal{N}}} f_{ij} \exp\Big(-\mathrm{i}\frac{2k\pi}{L_x}x_i\Big) \exp\Big(-\mathrm{i}\frac{2l\pi}{L_y}y_j\Big), \quad (k, l) \in \widehat{S}_{\mathcal{N}},$$

and f can be reconstructed by the inverse transform $f = P^{-1}\widehat{f}$ with components

$$f_{ij} = \sum_{(k,l)\in\widehat{S}_{\mathcal{N}}} \widehat{f}_{kl} \exp\Big(\mathrm{i}\frac{2k\pi}{L_x}x_i\Big) \exp\Big(\mathrm{i}\frac{2l\pi}{L_y}y_j\Big), \quad (i, j) \in S_{\mathcal{N}}.$$

Let $\widehat{\mathcal{M}}_{\mathcal{N}} = \{Pf \mid f \in \mathcal{M}_{\mathcal{N}}\}$ and define the operators $\widehat{D}_x$ and $\widehat{D}_y$ on $\widehat{\mathcal{M}}_{\mathcal{N}}$ as

$$(\widehat{D}_x\widehat{f})_{kl} = \Big(\frac{2k\pi\mathrm{i}}{L_x}\Big)\widehat{f}_{kl}, \quad (\widehat{D}_y\widehat{f})_{kl} = \Big(\frac{2k\pi\mathrm{i}}{L_y}\Big)\widehat{f}_{kl}, \quad (k, l) \in \widehat{S}_{\mathcal{N}}.$$

Then, the Fourier spectral approximations of the first- and second-order partial derivatives are given as

$$D_x = P^{-1}\widehat{D}_x P, \quad D_y = P^{-1}\widehat{D}_y P, \quad D_x^2 = P^{-1}\widehat{D}_x^2 P, \quad D_y^2 = P^{-1}\widehat{D}_y^2 P.$$

For any $f \in \mathcal{M}_{\mathcal{N}}$ and $\boldsymbol{f} = (f_1, f_2)^T \in \mathcal{M}_{\mathcal{N}} \times \mathcal{M}_{\mathcal{N}}$, the discrete gradient, divergence and Laplace operators are defined, respectively, by

$$\nabla_{\mathcal{N}} f = \begin{pmatrix} D_x f \\ D_y f \end{pmatrix}, \quad \nabla_{\mathcal{N}} \cdot \boldsymbol{f} = D_x f^1 + D_y f^2, \quad \Delta_{\mathcal{N}} f = D_x^2 f + D_y^2 f.$$

Let $g \in \mathcal{M}_{\mathcal{N}}$ and $\boldsymbol{g} = (g_1, g_2)^T \in \mathcal{M}_{\mathcal{N}} \times \mathcal{M}_{\mathcal{N}}$. The discrete L^2 inner product $\langle\cdot,\cdot\rangle$, the discrete L^2 norm $\|\cdot\|$, and the discrete L^∞ norm $\|\cdot\|_\infty$ are given as

$$\langle f, g\rangle = h_x h_y \sum_{(i,j)\in S_{\mathcal{N}}} f_{ij} g_{ij}, \quad \|f\| = \sqrt{\langle f, f\rangle}, \quad \|f\|_\infty = \max_{(i,j)\in S_{\mathcal{N}}} |f_{ij}|,$$

$$\langle \boldsymbol{f}, \boldsymbol{g} \rangle = h_x h_y \sum_{(i,j)\in S_{\mathcal{N}}} (f_{ij}^1 g_{ij}^1 + f_{ij}^2 g_{ij}^2), \quad \|\boldsymbol{f}\| = \sqrt{\langle \boldsymbol{f}, \boldsymbol{f} \rangle},$$
$$\|\boldsymbol{f}\|_\infty = \max_{(i,j)\in S_{\mathcal{N}}} \sqrt{|f_{ij}^1|^2 + |f_{ij}^2|^2}.$$

We also define $\mathcal{M}_{\mathcal{N}}^0$ as the zero-mean subspace of $\mathcal{M}_{\mathcal{N}}$, i.e.,

$$\mathcal{M}_{\mathcal{N}}^0 = \{ f \in \mathcal{M}_{\mathcal{N}} \mid \langle f, 1 \rangle = 0 \}.$$

Proposition 2 *For any functions $f, g \in \mathcal{M}_{\mathcal{N}}$ and $\boldsymbol{g} \in \mathcal{M}_{\mathcal{N}} \times \mathcal{M}_{\mathcal{N}}$, we have the following properties:*

(i) $\langle f, \nabla_{\mathcal{N}} \cdot \boldsymbol{g} \rangle = -\langle \nabla_{\mathcal{N}} f, \boldsymbol{g} \rangle$ *and* $\langle f, \Delta_{\mathcal{N}} g \rangle = -\langle \nabla_{\mathcal{N}} f, \nabla_{\mathcal{N}} g \rangle = \langle \Delta_{\mathcal{N}} f, g \rangle$;
(ii) $\Delta_{\mathcal{N}}$ *is non-positive definite on* $\mathcal{M}_{\mathcal{N}}$ *and negative definite on* $\mathcal{M}_{\mathcal{N}}^0$, *thus,* $\Delta_{\mathcal{N}}^{-1}$ *exists on* $\mathcal{M}_{\mathcal{N}}^0$.

We still use ϕ and ρ to express their numerical approximations on $\Omega_{\mathcal{N}}$ for simplicity. The spatially discretized version of the energy functional (1), $E : \mathcal{M}_{\mathcal{N}} \times \mathcal{M}_{\mathcal{N}} \to \mathbb{R}$, is defined as

$$E(\phi, \rho) = \frac{\eta_1}{2}\|\Delta_{\mathcal{N}}\phi\|^2 + \frac{\varepsilon}{2}\|\nabla_{\mathcal{N}}\phi\|^2 + \frac{\eta_2}{2}\|\nabla_{\mathcal{N}}\rho\|^2 + \Big\langle \frac{f(\phi)}{\varepsilon} + h(\rho) - \theta\rho|\nabla_{\mathcal{N}}\phi|^2,\ 1 \Big\rangle \quad (4)$$

Also, we define the discrete versions of G_c and G_e as $E_c, E_e : \mathcal{M}_{\mathcal{N}} \times \mathcal{M}_{\mathcal{N}} \to \mathbb{R}$, namely,

$$E_c(\phi, \rho) = \frac{\eta_1}{2}\|\Delta_{\mathcal{N}}\phi\|^2 + \frac{\varepsilon + A_2}{2}\|\nabla_{\mathcal{N}}\phi\|^2 + \frac{A_1}{2\varepsilon}\|\phi\|^2 + \frac{\eta_2}{2}\|\nabla_{\mathcal{N}}\rho\|^2 + \frac{B}{2}\|\rho\|^2,$$
$$E_e(\phi, \rho) = \Big\langle -\frac{f(\phi)}{\varepsilon} - h(\rho) + \theta\rho|\nabla_{\mathcal{N}}\phi|^2,\ 1 \Big\rangle + \frac{A_1}{2\varepsilon}\|\phi\|^2 + \frac{B}{2}\|\rho\|^2 + \frac{A_2}{2}\|\nabla_{\mathcal{N}}\phi\|^2$$

It is obvious that E_c is convex. For the convexity of E_e, we have the following result, the proof of which is similar to that of Proposition 1 and thus omitted for brevity.

Proposition 3 *Suppose that there exist positive constants $K_\phi \geq 1$, $\widetilde{K}_\phi > 0$ and $\frac{1}{2} < K_\rho < 1$ such that $\phi, \rho \in \mathcal{M}_{\mathcal{N}}$ satisfy $|\phi_{ij}| \leq K_\phi$, $|\nabla_{\mathcal{N}}\phi_{ij}| \leq \widetilde{K}_\phi$ and $1 - K_\rho \leq \rho_{ij} \leq K_\rho$ for any $1 \leq i, j \leq N$. If A_1, A_2 and B satisfy (3), the discrete energy $E_e(\phi, \rho)$ is convex with respect to ϕ and ρ.*

Based on the linear convex splitting of the discrete energy (4), we construct the space-discrete scheme for the BFS system (2) as follows: find $\phi, \rho : [0, T] \to \mathcal{M}_{\mathcal{N}}$ such that

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = \mathcal{L}_1\phi + F_1(\phi, \rho), \quad (5a)$$
$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \mathcal{L}_2\rho + F_2(\phi, \rho), \quad (5b)$$

where the linear operators $\mathcal{L}_1 = \eta_1\Delta_{\mathcal{N}}^3 - (\varepsilon + A_2)\Delta_{\mathcal{N}}^2 + \frac{A_1}{\varepsilon}\Delta_{\mathcal{N}}$ and $\mathcal{L}_2 = -\eta_2\Delta_{\mathcal{N}}^2 + B\Delta_{\mathcal{N}}$, and the nonlinear terms

$$F_1(\phi, \rho) = \Delta_{\mathcal{N}}\Big(\frac{f'(\phi)}{\varepsilon} - \frac{A_1}{\varepsilon}\phi + 2\theta\nabla_{\mathcal{N}} \cdot (\rho\nabla_{\mathcal{N}}\phi) + A_2\Delta_{\mathcal{N}}\phi\Big),$$
$$F_2(\phi, \rho) = \Delta_{\mathcal{N}}\big(h'(\rho) - B\rho - \theta|\nabla_{\mathcal{N}}\phi|^2\big).$$

Since the semi-discrete system (5) is the projection of the system (2) onto the finite-dimension space originated from the Fourier spectral approximations, the existence of a solution for the

system (5) is consistent with that of the system (2). Moreover, by similar analysis as in [32], we can obtain that the discrete energy (4) also satisfies the energy dissipation law, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}E(\phi(t),\rho(t)) \le 0.$$

Remark 2 By Proposition 2, we know that the linear operators $\mathcal{L}_1$ and $\mathcal{L}_2$ are symmetric and non-positive on $\mathcal{M}_{\mathcal{N}}$, i.e.,

$$\begin{aligned}
&\langle \mathcal{L}_1 u, v\rangle = \langle u, \mathcal{L}_1 v\rangle, \quad \langle \mathcal{L}_2 u, v\rangle = \langle u, \mathcal{L}_1 v\rangle,\\
&\langle \mathcal{L}_1 u, u\rangle = -\eta_1 \|\Delta_{\mathcal{N}}\nabla_{\mathcal{N}} u\|^2 - (\varepsilon + A_2)\|\Delta_{\mathcal{N}} u\|^2 - \frac{A_1}{\varepsilon}\|\nabla_{\mathcal{N}} u\|^2 \le 0,\\
&\langle \mathcal{L}_2 u, u\rangle = -\eta_2 \|\Delta_{\mathcal{N}} u\|^2 - B\|\nabla_{\mathcal{N}} u\|^2 \le 0,
\end{aligned}$$

for any $u, v \in \mathcal{M}_{\mathcal{N}}$. Moreover, both $\mathcal{L}_1$ and $\mathcal{L}_2$ are negative definite on $\mathcal{M}^0_{\mathcal{N}}$.

3.3 Exponential Time Differencing for Time Integration

For a given positive integer N_t, we partition the time interval $[0, T]$ by the nodes $t_n = n\tau$, $0 \le n \le N_t$, with time step $\tau = T/N_t$. By applying the variation-of-constants formula to the space-discrete system (5) and letting $t = t_{n+1}$, we obtain

$$\begin{aligned}
\phi(t_{n+1}) &= \mathrm{e}^{\tau\mathcal{L}_1}\phi(t_n) + \int_0^\tau \mathrm{e}^{(\tau-s)\mathcal{L}_1} F_1(\phi(t_n+s),\rho(t_n+s))\,\mathrm{d}s,\\
\rho(t_{n+1}) &= \mathrm{e}^{\tau\mathcal{L}_2}\rho(t_n) + \int_0^\tau \mathrm{e}^{(\tau-s)\mathcal{L}_2} F_2(\phi(t_n+s),\rho(t_n+s))\,\mathrm{d}s.
\end{aligned}$$

Denote by ϕ^n and ρ^n the numerical approximation of $\phi(t_n)$ and $\rho(t_n)$. Then, by the application of a third-order ETDRK (ETDRK3) scheme, we obtain the following fully-discrete numerical scheme for the system (2):

$$\phi_1 = \varphi_0(\tau\mathcal{L}_1)\phi^n + \tau\varphi_1(\tau\mathcal{L}_1)F_1(\phi^n,\rho^n), \tag{6a}$$

$$\rho_1 = \varphi_0(\tau\mathcal{L}_2)\rho^n + \tau\varphi_1(\tau\mathcal{L}_2)F_2(\phi^n,\rho^n), \tag{6b}$$

$$\phi_2 = \varphi_0(\tfrac{2}{3}\tau\mathcal{L}_1)\phi^n + \tau\widetilde{\varphi}_2(\tfrac{2}{3}\tau\mathcal{L}_1)F_1(\phi^n,\rho^n) + \tau\widetilde{\varphi}_3(\tfrac{2}{3}\tau\mathcal{L}_1)F_1(\phi_1,\rho_1), \tag{6c}$$

$$\rho_2 = \varphi_0(\tfrac{2}{3}\tau\mathcal{L}_2)\rho^n + \tau\widetilde{\varphi}_2(\tfrac{2}{3}\tau\mathcal{L}_2)F_2(\phi^n,\rho^n) + \tau\widetilde{\varphi}_3(\tfrac{2}{3}\tau\mathcal{L}_2)F_2(\phi_1,\rho_1), \tag{6d}$$

$$\begin{aligned}
\phi^{n+1} = {}& \varphi_0(\tau\mathcal{L}_1)\phi^n + \tau\widetilde{\varphi}_4(\tau\mathcal{L}_1)F_1(\phi^n,\rho^n) + \tau\widetilde{\varphi}_5(\tau\mathcal{L}_1)F_1(\phi_1,\rho_1)\\
&+ \tau\widetilde{\varphi}_6(\tau\mathcal{L}_1)F_1(\phi_2,\rho_2),
\end{aligned} \tag{6e}$$

$$\begin{aligned}
\rho^{n+1} = {}& \varphi_0(\tau\mathcal{L}_2)\rho^n + \tau\widetilde{\varphi}_4(\tau\mathcal{L}_2)F_2(\phi^n,\rho^n) + \tau\widetilde{\varphi}_5(\tau\mathcal{L}_2)F_2(\phi_1,\rho_1)\\
&+ \tau\widetilde{\varphi}_6(\tau\mathcal{L}_2)F_2(\phi_2,\rho_2),
\end{aligned} \tag{6f}$$

where $\varphi_0(a) = \mathrm{e}^a$, $\varphi_1(a) = (\mathrm{e}^a - 1)a^{-1}$, and $\varphi_2(a) = (\mathrm{e}^a - 1 - a)a^{-2}$ as defined in [20], and

$$\begin{aligned}
&\widetilde{\varphi}_2(a) = \frac{2}{3}\varphi_1(a) - \frac{4}{9}\varphi_2(a), \quad \widetilde{\varphi}_3(a) = \frac{4}{9}\varphi_2(a), \quad \widetilde{\varphi}_4(a) = \frac{3}{4}\varphi_1(a) - \varphi_2(a),\\
&\widetilde{\varphi}_5(a) = \varphi_2(a) - \frac{1}{2}\varphi_1(a), \quad \widetilde{\varphi}_6(a) = \frac{3}{4}\varphi_1(a).
\end{aligned}$$

Remark 3 Since both $\mathcal{L}_1$ and $\mathcal{L}_2$ are negative definite on $\mathcal{M}^0_{\mathcal{N}}$, we know that $\varphi_1(\tau\mathcal{L}_k)$ and $\widetilde{\varphi}_l(\tau\mathcal{L}_k)$ $(l = 2, 3, 4, 5, 6,\ k = 1, 2)$, which contain $\mathcal{L}_k^{-1}$, are well-defined only on $\mathcal{M}^0_{\mathcal{N}}$.

Moreover, we can obtain $F_1(\phi, \rho)$, $F_2(\phi, \rho) \in \mathcal{M}_{\mathcal{N}}^0$ for any $\phi, \rho \in \mathcal{M}_{\mathcal{N}}$ by their definitions due to the fact that $\Delta_{\mathcal{N}} u \in \mathcal{M}_{\mathcal{N}}^0$ for any $u \in \mathcal{M}_{\mathcal{N}}$. Thus, the ETDRK3 scheme (6) is well-defined.

Remark 4 In this paper, we consider the periodic boundary conditions for simplicity. The proposed scheme can be easily extended to the homogeneous Neumann boundary conditions, which is also a common choice in the simulation of the BFS phase field models. In this case, the spatial discretization can be implemented using the spectral collocation method based on the Chebyshev polynomials. The corresponding algorithm involves the discrete cosine transform, which can also be implemented efficiently by resorting to the fast Fourier transform. The analysis is similar to that of the periodic case, so we omit it for brevity.

By the similar spirit from [22], the ETDRK3 scheme (6) satisfies the following discrete version of the mass conservation held by the system (2).

Proposition 4 *The ETDRK3 scheme (6) is mass conservative in the discrete sense that, i.e., for* $0 \le n \le N_t - 1$,

$$\langle \phi^{n+1}, 1\rangle = \langle \phi_2, 1\rangle = \langle \phi_1, 1\rangle = \langle \phi^n, 1\rangle,$$
$$\langle \rho^{n+1}, 1\rangle = \langle \rho_2, 1\rangle = \langle \rho_1, 1\rangle = \langle \rho^n, 1\rangle.$$

For the subsequent analysis, we also need the following lemmas:

Lemma 1 (see [19]) *Let φ be defined on the spectrum of a symmetric matrix $L \in \mathbb{R}^{n\times n}$, that is, the values $\varphi(\lambda_i)$, $1 \le i \le n$, exist, where $\{\lambda_i\}_{i=1}^n$ are the eigenvalues of L. Then*

(i) *$\varphi(L)$ commutes with L;*
(ii) *$\varphi(L^\top) = \varphi(L)^\top$;*
(iii) *the eigenvalues of $\varphi(L)$ are $\{\varphi(\lambda_i) \mid 1 \le i \le n\}$;*
(iv) *$\varphi(P^{-1}LP) = P^{-1}\varphi(L)P$ for any nonsingular matrix $P \in \mathbb{C}^{n\times n}$;*
(v) *for any $P, Q \in \mathbb{C}^{n\times n}$, $\mathrm{e}^{(P+Q)t} = \mathrm{e}^{Pt}\mathrm{e}^{Qt} = \mathrm{e}^{Qt}\mathrm{e}^{Pt}$ if and only if $PQ = QP$;*
(vi) *$\frac{\mathrm{d}}{\mathrm{d}s}(\mathrm{e}^{sL}) = L\mathrm{e}^{sL} = \mathrm{e}^{sL}L$ for any $s \in \mathbb{R}$.*

Lemma 2 *(see [22]) Let $\mathcal{L} : \mathcal{M}_{\mathcal{N}} \to \mathcal{M}_{\mathcal{N}}$ be a linear and symmetric operator. Defining the norm of $\mathcal{L}$ as $|||\mathcal{L}||| = \max\{|\lambda| \mid \lambda \in \sigma(\mathcal{L})\}$ where $\sigma(\mathcal{L})$ is the spectrum of $\mathcal{L}$, it holds that*

$$\|\mathcal{L}v\| \le |||\mathcal{L}|||\|v\|, \quad \forall v \in \mathcal{M}_{\mathcal{N}}.$$

Lemma 3 *For any $a \le 0$, we have*

$$0 < \varphi_0(a) \le 1, \quad 0 < (1-a)\varphi_0(a) \le 1, \quad 1 \le (1-a)\varphi_1(a) < 2,$$
$$0 \le -a\varphi_1(a) < 1, \quad 0 < (1-a)\widetilde{\varphi}_2(a) < 1, \quad 0 \le -\frac{3a}{2}\widetilde{\varphi}_2(a) < 1,$$
$$0 < (1-a)\widetilde{\varphi}_3(a) < 1, \quad 0 \le -\frac{3a}{2}\widetilde{\varphi}_3(a) < 1, \quad -1 < (1-a)\widetilde{\varphi}_4(a) < 1,$$
$$0 \le (1-a)\widetilde{\varphi}_5(a) < 1, \quad 0 < (1-a)\widetilde{\varphi}_6(a) < 1.$$

From Lemmas 1 and 3, we can obtain, e.g., $|||(I - \tau\mathcal{L}_1)\varphi_0(\tau\mathcal{L}_1)||| \le 1$ since $\mathcal{L}_1$ is non-positive definite.

4 Analysis of Convergence and Bound-Preservation

Based on Proposition 3, we know that the boundedness assumption of ϕ^n, $|\nabla_{\mathcal{N}}\phi^n|$ and ρ^n are needed in order to choose appropriate stabilization parameters A_1, A_2 and B, which is essential to the proposed energy splitting. Since the numerical solution can be regarded as a combination of the exact solution and the error between the exact and numerical solutions, the bound-preserving property of the numerical solution (consequently the boundedness assumption always holds) will naturally be derived by assuming the regularity of the exact solution and ensuring that the numerical error is small enough. Thus, we mainly focus on the convergence analysis.

Let us denote by $(\phi(t), \rho(t))$ the solution of the space-discrete system (5). Plugging $(\phi(t_n), \rho(t_n))$ into (6), we have

$$\widetilde{\phi}_1 = \varphi_0(\tau\mathcal{L}_1)\phi(t_n) + \tau\varphi_1(\tau\mathcal{L}_1)F_1(\phi(t_n), \rho(t_n)), \tag{7a}$$

$$\widetilde{\rho}_1 = \varphi_0(\tau\mathcal{L}_2)\rho(t_n) + \tau\varphi_1(\tau\mathcal{L}_2)F_2(\phi(t_n), \rho(t_n)), \tag{7b}$$

$$\widetilde{\phi}_2 = \varphi_0(\tfrac{2}{3}\tau\mathcal{L}_1)\phi(t_n) + \tau\widetilde{\varphi}_2(\tfrac{2}{3}\tau\mathcal{L}_1)F_1(\phi(t_n), \rho(t_n) + \tau\widetilde{\varphi}_3(\tfrac{2}{3}\tau\mathcal{L}_1)F_1(\widetilde{\phi}_1, \widetilde{\rho}_1), \tag{7c}$$

$$\widetilde{\rho}_2 = \varphi_0(\tfrac{2}{3}\tau\mathcal{L}_2)\rho(t_n) + \tau\widetilde{\varphi}_2(\tfrac{2}{3}\tau\mathcal{L}_2)F_2(\phi(t_n), \rho(t_n)) + \tau\widetilde{\varphi}_3(\tfrac{2}{3}\tau\mathcal{L}_2)F_2(\widetilde{\phi}_1, \widetilde{\rho}_1), \tag{7d}$$

$$\begin{aligned}\phi(t_{n+1}) = {} & \varphi_0(\tau\mathcal{L}_1)\phi(t_n) + \tau\widetilde{\varphi}_4(\tau\mathcal{L}_1)F_1(\phi(t_n), \rho(t_n)) + \tau\widetilde{\varphi}_5(\tau\mathcal{L}_1)F_1(\widetilde{\phi}_1, \widetilde{\rho}_1) \\ & + \tau\widetilde{\varphi}_6(\tau\mathcal{L}_1)F_1(\widetilde{\phi}_2, \widetilde{\rho}_2) + R_n^{(1)},\end{aligned} \tag{7e}$$

$$\begin{aligned}\rho(t_{n+1}) = {} & \varphi_0(\tau\mathcal{L}_2)\rho(t_n) + \tau\widetilde{\varphi}_4(\tau\mathcal{L}_2)F_2(\phi(t_n), \rho(t_n)) + \tau\widetilde{\varphi}_5(\tau\mathcal{L}_2)F_2(\widetilde{\phi}_1, \widetilde{\rho}_1) \\ & + \tau\widetilde{\varphi}_6(\tau\mathcal{L}_2)F_2(\widetilde{\phi}_2, \widetilde{\rho}_2) + R_n^{(2)},\end{aligned} \tag{7f}$$

where $R_n^{(1)}$ and $R_n^{(2)}$ are truncation errors. Then we define

$$\begin{aligned} e_\phi^n &= \phi^{t_n} - \phi^n, \quad e_\rho^n = \rho^{t_n} - \rho^n, \\ e_{\phi_i} &= \widetilde{\phi}_i - \phi_i, \quad e_{\rho_i} = \widetilde{\rho}_i - \rho_i, \quad i = 1, 2,\end{aligned}$$

where we use ϕ^{t_n} and ρ^{t_n} to express $\phi(t_n)$ and $\rho(t_n)$ for simplicity. Subtracting (6) from (7), we get

$$e_{\phi_1} = \varphi_0(\tau\mathcal{L}_1)e_\phi^n + \tau\varphi_1(\tau\mathcal{L}_1)[F_1(\phi^{t_n}, \rho^{t_n}) - F_1(\phi^n, \rho^n)], \tag{8a}$$

$$e_{\rho_1} = \varphi_0(\tau\mathcal{L}_2)e_\rho^n + \tau\varphi_1(\tau\mathcal{L}_2)[F_2(\phi^{t_n}, \rho^{t_n}) - F_2(\phi^n, \rho^n)], \tag{8b}$$

$$\begin{aligned} e_{\phi_2} = {} & \varphi_0(\tfrac{2}{3}\tau\mathcal{L}_1)e_\phi^n + \tau\widetilde{\varphi}_2(\tfrac{2}{3}\tau\mathcal{L}_1)[F_1(\phi^{t_n}, \rho^{t_n}) - F_1(\phi^n, \rho^n)] \\ & + \tau\widetilde{\varphi}_3(\tfrac{2}{3}\tau\mathcal{L}_1)[F_1(\widetilde{\phi}_1, \widetilde{\rho}_1) - F_1(\phi_1, \rho_1)],\end{aligned} \tag{8c}$$

$$\begin{aligned} e_{\rho_2} = {} & \varphi_0(\tfrac{2}{3}\tau\mathcal{L}_2)e_\rho^n + \tau\widetilde{\varphi}_2(\tfrac{2}{3}\tau\mathcal{L}_2)[F_2(\phi^{t_n}, \rho^{t_n}) - F_2(\phi^n, \rho^n)] \\ & + \tau\widetilde{\varphi}_3(\tfrac{2}{3}\tau\mathcal{L}_2)[F_2(\widetilde{\phi}_1, \widetilde{\rho}_1) - F_2(\phi_1, \rho_1)],\end{aligned} \tag{8d}$$

$$\begin{aligned} e_\phi^{n+1} = {} & \varphi_0(\tau\mathcal{L}_1)e_\phi^n + \tau\widetilde{\varphi}_4(\tau\mathcal{L}_1)[F_1(\phi^{t_n}, \rho^{t_n}) - F_1(\phi^n, \rho^n)] \\ & + \tau\widetilde{\varphi}_5(\tau\mathcal{L}_1)[F_1(\widetilde{\phi}_1, \widetilde{\rho}_1) - F_1(\phi_1, \rho_1)] \\ & + \tau\widetilde{\varphi}_6(\tau\mathcal{L}_1)[F_1(\widetilde{\phi}_2, \widetilde{\rho}_2) - F_1(\phi_2, \rho_2)] + R_n^{(1)},\end{aligned} \tag{8e}$$

$$\begin{aligned} e_\rho^{n+1} = {} & \varphi_0(\tau\mathcal{L}_2)e_\rho^n + \tau\widetilde{\varphi}_4(\tau\mathcal{L}_2)[F_2(\phi^{t_n}, \rho^{t_n}) - F_2(\phi^n, \rho^n)] \\ & + \tau\widetilde{\varphi}_5(\tau\mathcal{L}_2)[F_2(\widetilde{\phi}_1, \widetilde{\rho}_1) - F_2(\phi_1, \rho_1)] \\ & + \tau\widetilde{\varphi}_6(\tau\mathcal{L}_2)[F_2(\widetilde{\phi}_2, \widetilde{\rho}_2) - F_2(\phi_2, \rho_2)] + R_n^{(2)}.\end{aligned} \tag{8f}$$

Assuming that $(\phi(t), \rho(t))$ is smooth enough on $\overline{\Omega} \times [0, T]$, there exist constants $M_0 \geq 1$, $\widetilde{M}_0 \geq 0$ and $0 < \delta_0 < 1$ such that for $1 \leq i \leq N_x$, $1 \leq j \leq N_y$ and $t \in [0, T]$,

$$|\phi_{ij}(t)| \leq M_0, \quad |\nabla_{\mathcal{N}} \phi_{ij}(t)| \leq \widetilde{M}_0, \quad \delta_0 \leq \rho_{ij}(t) \leq 1 - \delta_0. \tag{9}$$

Denoting $K_\phi = M_0 + 1$, $\widetilde{K}_\phi = \widetilde{M}_0 + 1$, and $K_\rho = 1 - \delta_0/2$, we will prove that under appropriate conditions, ϕ^n, $|\nabla_{\mathcal{N}} \phi^n|$ and ρ^n are uniformly bounded by K_ϕ, $\widetilde{K}_\phi$ and K_ρ respectively.

Theorem 1 *Suppose that $\phi(t)$ and $\rho(t)$ are smooth enough such that $\Delta_{\mathcal{N}}^5 \phi \in H^3(0, T; \mathcal{M}_{\mathcal{N}})$, and $\nabla_{\mathcal{N}} \Delta_{\mathcal{N}}^4 \rho \in H^3(0, T; \mathcal{M}_{\mathcal{N}})$. There exists some fixed constant $C_0 > 0$, such that if h_x, h_y and τ are sufficiently small subject to a constraint $\tau \leq \min\{1, C_0 h_x^{\frac{1}{2}} h_y^{\frac{1}{2}}\}$, then the ETDRK3 scheme (6) satisfies the error estimate*

$$\|\phi^n - \phi(t_n)\| + \|\rho^n - \rho(t_n)\| \leq C\tau^3, \quad 0 \leq n \leq N_t, \tag{10}$$

and the bound-preserving property: for $1 \leq i \leq N_x$, $1 \leq j \leq N_y$, $0 \leq n \leq N_t$,

$$|\phi_{ij}^n| \leq K_\phi, \quad |\nabla_{\mathcal{N}} \phi_{ij}^n| \leq \widetilde{K}_\phi, \quad 1 - K_\rho \leq \rho_{ij}^n \leq K_\rho, \tag{11}$$

where $C > 0$ is a constant independent of h_x, h_y and τ.

Proof We prove this theorem by induction. Assume that (10) and (11) have been obtained for $k = 0, 1, \ldots, n$ $(0 < n < N_t)$, we then prove that ϕ^{n+1} and ρ^{n+1} satisfy (10) and (11).

Note that the regularity assumption for ϕ and ρ implies that (9) holds. By assuming the sufficient smoothness of $\widetilde{\phi}_i$, $\widetilde{\rho}_i$ and $\nabla_{\mathcal{N}} \widetilde{\phi}_i$, $i = 1, 2$, we know that they satisfy (9) as well. The boundedness of ϕ_i, ρ_i and $|\nabla_{\mathcal{N}} \phi_i|$, $i = 1, 2$ at the $(n+1)$-th step can be proved in a similar way as that of ϕ^{n+1}, ρ^{n+1} and $|\nabla_{\mathcal{N}} \phi^{n+1}|$ in the subsequent analysis, thus we omit its proof for brevity.

To make the proof more readable, we organize it by four parts.

Part 1. Beginning with the error equations (8e) and (8f), we will derive (23) as an estimate for e_ϕ^{n+1} and e_ρ^{n+1}.

To this end, we act $(I - \tau\mathcal{L}_1)$ on both sides of (8e) and take the discrete L^2 inner product on the derived equation with e_ϕ^{n+1} to give

$$\begin{aligned}
&\|e_\phi^{n+1}\|^2 + \frac{A_1}{\varepsilon}\tau\|\nabla_{\mathcal{N}} e_\phi^{n+1}\|^2 + (\varepsilon + A_2)\tau\|\Delta_{\mathcal{N}} e_\phi^{n+1}\|^2 + \eta_1\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_\phi^{n+1}\|^2 \\
&\quad = \left\langle (I - \tau\mathcal{L}_1)\varphi_0(\tau\mathcal{L}_1) e_\phi^n, e_\phi^{n+1} \right\rangle + \left\langle (I - \tau\mathcal{L}_1) R_n^{(1)}, e_\phi^{n+1} \right\rangle \\
&\qquad + \tau\left\langle (I - \tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)[F_1(\phi^{t_n}, \rho^{t_n}) - F_1(\phi^n, \rho^n)], e_\phi^{n+1} \right\rangle \\
&\qquad + \tau\left\langle (I - \tau\mathcal{L}_1)\widetilde{\varphi}_5(\tau\mathcal{L}_1)[F_1(\widetilde{\phi}_1, \widetilde{\rho}_1) - F_1(\phi_1, \rho_1)], e_\phi^{n+1} \right\rangle \\
&\qquad + \tau\left\langle (I - \tau\mathcal{L}_1)\widetilde{\varphi}_6(\tau\mathcal{L}_1)[F_1(\widetilde{\phi}_2, \widetilde{\rho}_2) - F_1(\phi_2, \rho_2)], e_\phi^{n+1} \right\rangle \\
&\quad =: I_{11} + I_{12} + I_{13} + I_{14} + I_{15}.
\end{aligned} \tag{12}$$

By the application of Lemmas 1, 2 and 3, we can obtain

$$I_{11} \leq \|(I - \tau\mathcal{L}_1)\varphi_0(\tau\mathcal{L}_1)\| \|e_\phi^n\| \|e_\phi^{n+1}\| \leq \frac{1}{2}\|e_\phi^n\|^2 + \frac{1}{2}\|e_\phi^{n+1}\|^2. \tag{13}$$

Using similar analysis as in [22, Lemma 4.4], we have

$$\|(I - \tau\mathcal{L}_1) R_n^{(1)}\| \leq C_1\tau^4, \quad \|(I - \tau\mathcal{L}_2) R_n^{(2)}\| \leq C_2\tau^4,$$

where C_1 and C_2 are positive constants depending on η_1, η_2, ε, θ, β, β_1, A_1, A_2, B, T, Ω and the regularity of the exact solution of the system (2), but independent of h_x, h_y and τ. Thus,

$$I_{12} \le \|(I-\tau\mathcal{L}_1)R_n^{(1)}\|\|e_\phi^{n+1}\| \le C_1^2\tau^7 + \frac{\tau}{4}\|e_\phi^{n+1}\|^2. \tag{14}$$

Then, we rewrite I_{13} as

$$\begin{aligned} I_{13} &= \frac{\tau}{\varepsilon}\big\langle (I-\tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\Delta_{\mathcal{N}}\big(f'(\phi^{t_n})-f'(\phi^n)\big), e_\phi^{n+1}\big\rangle \\ &\quad - \frac{A_1}{\varepsilon}\tau\big\langle (I-\tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\Delta_{\mathcal{N}}(\phi^{t_n}-\phi^n), e_\phi^{n+1}\big\rangle \\ &\quad + 2\theta\tau\big\langle (I-\tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\Delta_{\mathcal{N}}[\nabla_{\mathcal{N}}\cdot(\rho^{t_n}\nabla_{\mathcal{N}}\phi^{t_n}) \\ &\qquad - \nabla_{\mathcal{N}}\cdot(\rho^n\nabla_{\mathcal{N}}\phi^n)], e_\phi^{n+1}\big\rangle \\ &\quad + A_2\tau\big\langle (I-\tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\Delta_{\mathcal{N}}^2(\phi^{t_n}-\phi^n), e_\phi^{n+1}\big\rangle \\ &=: I_{131}+I_{132}+I_{133}+I_{134}. \end{aligned} \tag{15}$$

Since $(I-\tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)$ commutes with $\Delta_{\mathcal{N}}$ and $\|(I-\tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\| < 1$, we obtain

$$\begin{aligned} I_{131}+I_{132} &\le \frac{\tau}{\varepsilon}\|f'(\phi^{t_n})-f'(\phi^n)\|\|\Delta_{\mathcal{N}}e_\phi^{n+1}\| + \frac{A_1}{\varepsilon}\tau\|\phi^{t_n}-\phi^n\|\|\Delta_{\mathcal{N}}e_\phi^{n+1}\| \\ &\le \frac{K_f}{\varepsilon}\tau\|e_\phi^n\|\|\Delta_{\mathcal{N}}e_\phi^{n+1}\| + \frac{A_1}{\varepsilon}\tau\|e_\phi^n\|\|\Delta_{\mathcal{N}}e_\phi^{n+1}\| \\ &\le \frac{3(K_f+A_1)^2}{2\varepsilon^2A_2}\tau\|e_\phi^n\|^2 + \frac{A_2}{6}\tau\|\Delta_{\mathcal{N}}e_\phi^{n+1}\|^2, \end{aligned} \tag{16}$$

where we use the facts that $\|\phi^{t_n}\|_\infty \le M_0 < K_\phi$, $\|\phi^n\|_\infty \le K_\phi$, and $f''(\xi) = 3\xi^2 - 1 \le 3K_\phi^2 - 1 =: K_f$ for any $-K_\phi \le \xi \le K_\phi$. Then, the term I_{133} can be estimated as follows:

$$\begin{aligned} I_{133} &= -2\theta\tau\big\langle (I-\tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)(\rho^{t_n}\nabla_{\mathcal{N}}\phi^{t_n}-\rho^{t_n}\nabla_{\mathcal{N}}\phi^n), \nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^{n+1}\big\rangle \\ &\quad - 2\theta\tau\big\langle (I-\tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)(\rho^{t_n}\nabla_{\mathcal{N}}\phi^n-\rho^n\nabla_{\mathcal{N}}\phi^n), \nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^{n+1}\big\rangle \\ &\le 2\theta\tau\big(\|\rho^{t_n}\|_\infty\|\nabla_{\mathcal{N}}e_\phi^n\|\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^{n+1}\| + \|\nabla_{\mathcal{N}}\phi^n\|_\infty\|e_\rho^n\|\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^{n+1}\|\big) \\ &\le 2\theta\tau\|\nabla_{\mathcal{N}}e_\phi^n\|\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^{n+1}\| + 2\theta\widetilde{K}_\phi\tau\|e_\rho^n\|\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^{n+1}\| \\ &\le \frac{24\theta^2}{\eta_1}\tau\|\nabla_{\mathcal{N}}e_\phi^n\|^2 + \frac{24\theta^2\widetilde{K}_\phi^2}{\eta_1}\tau\|e_\rho^n\|^2 + \frac{\eta_1}{12}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^{n+1}\|^2. \end{aligned} \tag{17}$$

The term I_{134} can be addressed similarly as I_{132}, i.e.,

$$I_{134} \le A_2\tau\|\Delta_{\mathcal{N}}e_\phi^n\|\|\Delta_{\mathcal{N}}e_\phi^{n+1}\| \le \frac{3A_2}{2}\tau\|\Delta_{\mathcal{N}}e_\phi^n\|^2 + \frac{A_2}{6}\tau\|\Delta_{\mathcal{N}}e_\phi^{n+1}\|^2. \tag{18}$$

A combination of (13), (14), (15), (16), (17) and (18) results in

$$\begin{aligned} &I_{11}+I_{12}+I_{13} \\ &\le \frac{1}{2}\|e_\phi^n\|^2 + \frac{1}{2}\|e_\phi^{n+1}\|^2 + C_1^2\tau^7 + \frac{\tau}{4}\|e_\phi^{n+1}\|^2 + \frac{3(K_f+A_1)^2}{2\varepsilon^2A_2}\tau\|e_\phi^n\|^2 \\ &\quad + \frac{24\theta^2\widetilde{K}_\phi^2}{\eta_1}\tau\|e_\rho^n\|^2 + \frac{24\theta^2}{\eta_1}\tau\|\nabla_{\mathcal{N}}e_\phi^n\|^2 + \frac{3A_2}{2}\tau\|\Delta_{\mathcal{N}}e_\phi^n\|^2 \end{aligned}$$

$$+\frac{A_2}{3}\tau\|\Delta_{\mathcal{N}}e_{\phi}^{n+1}\|^2+\frac{\eta_1}{12}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi}^{n+1}\|^2. \tag{19}$$

Similarly, we can obtain the estimate for I_{14} and I_{15}, derived from the bound for I_{13} with e_{ϕ}^n replaced by e_{ϕ_1} and e_{ϕ_2}, and e_{ρ}^n replaced by e_{ρ_1} and e_{ρ_2}, respectively. Then, acting $(I-\tau\mathcal{L}_2)$ on both sides of (8f) and taking the discrete L^2 inner product on the derived equation with e_{ρ}^{n+1} lead to

$$\begin{aligned}
&\|e_{\rho}^{n+1}\|^2+B\tau\|\nabla_{\mathcal{N}}e_{\rho}^{n+1}\|^2+\eta_2\tau\|\Delta_{\mathcal{N}}e_{\rho}^{n+1}\|^2\\
&\quad=\left\langle(I-\tau\mathcal{L}_2)\varphi_0(\tau\mathcal{L}_2)e_{\rho}^n,e_{\rho}^{n+1}\right\rangle+\left\langle(I-\tau\mathcal{L}_2)R_n^{(2)},e_{\rho}^{n+1}\right\rangle\\
&\qquad+\tau\left\langle(I-\tau\mathcal{L}_2)\widetilde{\varphi}_4(\tau\mathcal{L}_2)[F_2(\phi^{t_n},\rho^{t_n})-F_2(\phi^n,\rho^n)],e_{\rho}^{n+1}\right\rangle\\
&\qquad+\tau\left\langle(I-\tau\mathcal{L}_2)\widetilde{\varphi}_5(\tau\mathcal{L}_2)[F_2(\widetilde{\phi}_1,\widetilde{\rho}_1)-F_2(\phi_1,\rho_1)],e_{\rho}^{n+1}\right\rangle\\
&\qquad+\tau\left\langle(I-\tau\mathcal{L}_2)\widetilde{\varphi}_6(\tau\mathcal{L}_2)[F_2(\widetilde{\phi}_2,\widetilde{\rho}_2)-F_2(\phi_2,\rho_2)],e_{\rho}^{n+1}\right\rangle\\
&\quad=:I_{21}+I_{22}+I_{23}+I_{24}+I_{25}.
\end{aligned} \tag{20}$$

By similar deductions as I_{11}, I_{12} and I_{13}, we get

$$I_{21}\le\frac{1}{2}\|e_{\rho}^n\|^2+\frac{1}{2}\|e_{\rho}^{n+1}\|^2,\quad I_{22}\le C_2^2\tau^7+\frac{\tau}{4}\|e_{\rho}^{n+1}\|^2, \tag{21}$$

$$\begin{aligned}
I_{23}&=\tau\left\langle(I-\tau\mathcal{L}_2)\widetilde{\varphi}_4(\tau\mathcal{L}_2)\Delta_{\mathcal{N}}\big(h'(\rho^{t_n})-h'(\rho^n)\big),e_{\rho}^{n+1}\right\rangle\\
&\quad-B\tau\left\langle(I-\tau\mathcal{L}_2)\widetilde{\varphi}_4(\tau\mathcal{L}_2)\Delta_{\mathcal{N}}(\rho^{t_n}-\rho^n),e_{\rho}^{n+1}\right\rangle\\
&\quad-\theta\tau\left\langle(I-\tau\mathcal{L}_2)\widetilde{\varphi}_4(\tau\mathcal{L}_2)\Delta_{\mathcal{N}}(|\nabla_{\mathcal{N}}\phi^{t_n}|^2-|\nabla_{\mathcal{N}}\phi^n|^2),e_{\rho}^{n+1}\right\rangle\\
&\le\frac{3(K_h+B)^2}{2\eta_2}\tau\|e_{\rho}^n\|^2+\frac{6\theta^2\widetilde{K}_{\phi}^2}{\eta_2}\tau\|\nabla_{\mathcal{N}}e_{\phi}^n\|^2+\frac{\eta_2}{3}\tau\|\Delta_{\mathcal{N}}e_{\rho}^{n+1}\|^2
\end{aligned} \tag{22}$$

due to the fact that there exists K_h such that

$$h''(\xi)=\frac{\beta_1}{\xi(1-\xi)}-\beta_2\le K_h,$$

for any $1-K_{\rho}\le\xi\le K_{\rho}$, and

$$\||\nabla_{\mathcal{N}}\phi^{t_n}|^2-|\nabla_{\mathcal{N}}\phi^n|^2\|=\|(\nabla_{\mathcal{N}}\phi^{t_n}+\nabla_{\mathcal{N}}\phi^n)\nabla_{\mathcal{N}}e_{\phi}^n\|\le 2\widetilde{K}_{\phi}\|\nabla_{\mathcal{N}}e_{\phi}^n\|.$$

Similarly, we can obtain the estimate for I_{24} and I_{25}, given by the bound for I_{23} with e_{ϕ}^n replaced by e_{ϕ_1} and e_{ϕ_2}, and e_{ρ}^n replaced by e_{ρ_1} and e_{ρ_2}, respectively. Combining the resulted inequalities with (12), (19), (20) (21) and (22) leads to

$$\begin{aligned}
&\frac{1}{2}\|e_{\phi}^{n+1}\|^2+\frac{1}{2}\|e_{\rho}^{n+1}\|^2+\frac{A_1}{\varepsilon}\tau\|\nabla_{\mathcal{N}}e_{\phi}^{n+1}\|^2+B\tau\|\nabla_{\mathcal{N}}e_{\rho}^{n+1}\|^2\\
&\quad+\varepsilon\tau\|\Delta_{\mathcal{N}}e_{\phi}^{n+1}\|^2+\frac{3\eta_1}{4}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi}^{n+1}\|^2,\\
&\le\frac{1}{2}\|e_{\phi}^n\|^2+\frac{1}{2}\|e_{\rho}^n\|^2+\frac{\tau}{4}\|e_{\phi}^{n+1}\|^2+\frac{\tau}{4}\|e_{\rho}^{n+1}\|^2+(C_1^2+C_2^2)\tau^7+I_3+I_4+I_5,
\end{aligned} \tag{23}$$

where

$$I_3:=\tfrac{3(K_f+A_1)^2}{2\varepsilon^2A_2}\tau\|e_{\phi}^n\|^2+K_1\tau\|e_{\rho}^n\|^2+K_2\tau\|\nabla_{\mathcal{N}}e_{\phi}^n\|^2+\tfrac{3A_2}{2}\tau\|\Delta_{\mathcal{N}}e_{\phi}^n\|^2, \tag{24}$$

$$I_4:=\tfrac{3(K_f+A_1)^2}{2\varepsilon^2A_2}\tau\|e_{\phi_1}\|^2+K_1\tau\|e_{\rho_1}\|^2+K_2\tau\|\nabla_{\mathcal{N}}e_{\phi_1}\|^2+\tfrac{3A_2}{2}\tau\|\Delta_{\mathcal{N}}e_{\phi_1}\|^2, \tag{25}$$

$$I_5 := \tfrac{3(K_f+A_1)^2}{2\varepsilon^2 A_2}\tau\|e_{\phi_2}\|^2 + K_1\tau\|e_{\rho_2}\|^2 + K_2\tau\|\nabla_{\mathcal{N}} e_{\phi_2}\|^2 + \tfrac{3A_2}{2}\tau\|\Delta_{\mathcal{N}} e_{\phi_2}\|^2, \tag{26}$$

with $K_1 := \frac{24\theta^2\widetilde{K}_\phi^2}{\eta_1} + \frac{3(K_h+B)^2}{2\eta_2}$ and $K_2 := \frac{24\theta^2}{\eta_1} + \frac{6\theta^2\widetilde{K}_\phi^2}{\eta_2}$. We estimate the last two terms in I_5 as follows:

$$\begin{aligned}
&K_2\tau\|\nabla_{\mathcal{N}} e_{\phi_2}\|^2 + \frac{3A_2}{2}\tau\|\Delta_{\mathcal{N}} e_{\phi_2}\|^2 \\
&\le \Big(K_2 + \frac{9A_2^2}{2\eta_1}\Big)\tau\|\nabla_{\mathcal{N}} e_{\phi_2}\|^2 + \frac{\eta_1}{8}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_{\phi_2}\|^2 \\
&\le \frac{3}{8\varepsilon}\Big(K_2 + \frac{9A_2^2}{2\eta_1}\Big)^2\tau\|e_{\phi_2}\|^2 + \frac{2\varepsilon}{3}\tau\|\Delta_{\mathcal{N}} e_{\phi_2}\|^2 + \frac{\eta_1}{8}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_{\phi_2}\|^2,
\end{aligned} \tag{27}$$

so that

$$I_5 \le K_3\tau\|e_{\phi_2}\|^2 + K_3\tau\|e_{\rho_2}\|^2 + \frac{2\varepsilon}{3}\tau\|\Delta_{\mathcal{N}} e_{\phi_2}\|^2 + \frac{\eta_1}{8}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_{\phi_2}\|^2, \tag{28}$$

where $K_3 := \max\big\{\frac{3(K_f+A_1)^2}{2\varepsilon^2 A_2} + \frac{3}{8\varepsilon}\big(K_2 + \frac{9A_2^2}{2\eta_1}\big)^2, K_1\big\}$.

Part 2. To obtain an estimate for the right-hand side of (28), we turn to the error equations (8c) and (8d), which leads to (32), the estimate for $I_4 + I_5$.

Acting $(I - \frac{2}{3}\tau\mathcal{L}_1)$ and $(I - \frac{2}{3}\tau\mathcal{L}_2)$ on both sides of (8c) and (8d) respectively, taking the discrete L^2 inner product on the first resulted equation with e_{ϕ_2}, the second resulted equation with e_{ρ_2}, and deducing similarly as for (8e) and (8f) lead to

$$\begin{aligned}
&\|e_{\phi_2}\|^2 + \|e_{\rho_2}\|^2 \\
&\le \|e_\phi^n\|^2 + \|e_\rho^n\|^2 + \frac{3(K_f+A_1)^2}{\varepsilon^2 A_2}\tau\|e_\phi^n\|^2 + \Big(\frac{12\theta^2}{\eta_1} + \frac{12\theta^2\widetilde{K}_\phi^2}{\eta_2}\Big)\tau\|\nabla_{\mathcal{N}} e_\phi^n\|^2 \\
&\quad + \Big(\frac{12\theta^2\widetilde{K}_\phi^2}{\eta_1} + \frac{3(K_h+B)^2}{\eta_2}\Big)\tau\|e_\rho^n\|^2 + 3A_2\tau\|\Delta_{\mathcal{N}} e_\phi^n\|^2 \\
&\quad + \frac{3(K_f+A_1)^2}{\varepsilon^2 A_2}\tau\|e_{\phi_1}\|^2 + \Big(\frac{12\theta^2}{\eta_1} + \frac{12\theta^2\widetilde{K}_\phi^2}{\eta_2}\Big)\tau\|\nabla_{\mathcal{N}} e_{\phi_1}\|^2 \\
&\quad + \Big(\frac{12\theta^2\widetilde{K}_\phi^2}{\eta_1} + \frac{3(K_h+B)^2}{\eta_2}\Big)\tau\|e_{\rho_1}\|^2 + 3A_2\tau\|\Delta_{\mathcal{N}} e_{\phi_1}\|^2,
\end{aligned} \tag{29}$$

which gives the estimate for the first two terms in (28). Since $\|\varphi_0(\frac{2}{3}\tau\mathcal{L}_1)\| \le 1$, $\|-\tau\mathcal{L}_1\widetilde{\varphi}_2(\frac{2}{3}\tau\mathcal{L}_1)\| < 1$ and $\|-\tau\mathcal{L}_1\widetilde{\varphi}_3(\frac{2}{3}\tau\mathcal{L}_1)\| < 1$, acting $-\tau\mathcal{L}_1$ on both sides of (8c) and taking the discrete L^2 inner product on the resulted equation with e_{ϕ_2}, we have

$$\begin{aligned}
&\frac{A_1}{\varepsilon}\tau\|\nabla_{\mathcal{N}} e_{\phi_2}\|^2 + (\varepsilon + A_2)\tau\|\Delta_{\mathcal{N}} e_{\phi_2}\|^2 + \eta_1\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_{\phi_2}\|^2 \\
&= \big\langle -\tau\mathcal{L}_1\varphi_0(\tfrac{2}{3}\tau\mathcal{L}_1)e_\phi^n, e_{\phi_2}\big\rangle \\
&\quad + \tau\big\langle -\tau\mathcal{L}_1\widetilde{\varphi}_2(\tfrac{2}{3}\tau\mathcal{L}_1)[F_1(\phi^{t_n},\rho^{t_n}) - F_1(\phi^n,\rho^n)], e_{\phi_2}\big\rangle \\
&\quad + \tau\big\langle -\tau\mathcal{L}_1\widetilde{\varphi}_3(\tfrac{2}{3}\tau\mathcal{L}_1)[F_1(\widetilde{\phi}_1,\widetilde{\rho}_1) - F_1(\phi_1,\rho_1)], e_{\phi_2}\big\rangle \\
&\le \frac{A_1}{4\varepsilon}\tau\|\nabla_{\mathcal{N}} e_\phi^n\|^2 + \frac{A_1}{\varepsilon}\tau\|\nabla_{\mathcal{N}} e_{\phi_2}\|^2 + \frac{3(\varepsilon+A_2)}{4}\tau\|\Delta_{\mathcal{N}} e_\phi^n\|^2 \\
&\quad + \frac{\varepsilon+A_2}{3}\tau\|\Delta_{\mathcal{N}} e_{\phi_2}\|^2 + \frac{\eta_1}{3}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_\phi^n\|^2 + \frac{3\eta_1}{4}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_{\phi_2}\|^2
\end{aligned}$$

$$
\begin{aligned}
&+\frac{3(K_f+A_1)^2}{2\varepsilon^2 A_2}\tau\|e_\phi^n\|^2+\frac{3A_2}{2}\tau\|\Delta_{\mathcal{N}}e_\phi^n\|^2+\frac{A_2}{3}\|\Delta_{\mathcal{N}}e_{\phi_2}\|^2\\
&+\frac{32\theta^2}{\eta_1}\tau\|\nabla_{\mathcal{N}}e_\phi^n\|^2+\frac{32\theta^2\widetilde{K}_\phi^2}{\eta_1}\tau\|e_\rho^n\|^2+\frac{\eta_1}{16}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi_2}\|^2\\
&+\frac{3(K_f+A_1)^2}{2\varepsilon^2 A_2}\tau\|e_{\phi_1}\|^2+\frac{3A_2}{2}\tau\|\Delta_{\mathcal{N}}e_{\phi_1}\|^2+\frac{A_2}{3}\|\Delta_{\mathcal{N}}e_{\phi_2}\|^2\\
&+\frac{32\theta^2}{\eta_1}\tau\|\nabla_{\mathcal{N}}e_{\phi_1}\|^2+\frac{32\theta^2\widetilde{K}_\phi^2}{\eta_1}\tau\|e_{\rho_1}\|^2+\frac{\eta_1}{16}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi_2}\|^2.
\end{aligned}
$$

Then, we obtain the estimate for the last two terms in (28) as follows:

$$
\begin{aligned}
&\frac{2\varepsilon}{3}\tau\|\Delta_{\mathcal{N}}e_{\phi_2}\|^2+\frac{\eta_1}{8}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi_2}\|^2\\
&\le\frac{A_1}{4\varepsilon}\tau\|\nabla_{\mathcal{N}}e_\phi^n\|^2+\frac{3(\varepsilon+A_2)}{4}\tau\|\Delta_{\mathcal{N}}e_\phi^n\|^2+\frac{\eta_1}{3}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^n\|^2\\
&\quad+\frac{3(K_f+A_1)^2}{2\varepsilon^2 A_2}\tau\|e_\phi^n\|^2+\frac{32\theta^2\widetilde{K}_\phi^2}{\eta_1}\tau\|e_\rho^n\|^2+\frac{32\theta^2}{\eta_1}\tau\|\nabla_{\mathcal{N}}e_\phi^n\|^2\\
&\quad+\frac{3A_2}{2}\tau\|\Delta_{\mathcal{N}}e_\phi^n\|^2+\frac{3(K_f+A_1)^2}{2\varepsilon^2 A_2}\tau\|e_{\phi_1}\|^2+\frac{32\theta^2\widetilde{K}_\phi^2}{\eta_1}\tau\|e_{\rho_1}\|^2\\
&\quad+\frac{32\theta^2}{\eta_1}\tau\|\nabla_{\mathcal{N}}e_{\phi_1}\|^2+\frac{3A_2}{2}\tau\|\Delta_{\mathcal{N}}e_{\phi_1}\|^2.
\end{aligned}
\tag{30}
$$

If $\tau<1$, a combination of (28), (29) and (30) yields

$$
\begin{aligned}
I_5\le{}&K_4\tau\|e_\phi^n\|^2+K_5\tau\|e_\rho^n\|^2+K_6\tau\|\nabla_{\mathcal{N}}e_\phi^n\|^2+K_7\tau\|\Delta_{\mathcal{N}}e_\phi^n\|_{\mathcal{N}}^2\\
&+\frac{\eta_1}{3}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_\phi^n\|^2+K_8\tau\|e_{\phi_1}\|^2+K_9\tau\|e_{\rho_1}\|^2+K_{10}\tau\|\nabla_{\mathcal{N}}e_{\phi_1}\|^2\\
&+K_{11}\tau\|\Delta_{\mathcal{N}}e_{\phi_1}\|^2,
\end{aligned}
\tag{31}
$$

where

$$
\begin{aligned}
K_4&:=K_3+K_3\tfrac{3(K_f+A_1)^2}{\varepsilon^2A_2}+\tfrac{3(K_f+A_1)^2}{2\varepsilon^2A_2},\\
K_5&:=K_3+K_3\Big(\tfrac{12\theta^2\widetilde{K}_\phi^2}{\eta_1}+\tfrac{3(K_h+B)^2}{\eta_2}\Big)+\tfrac{32\theta^2\widetilde{K}_\phi^2}{\eta_1},\\
K_6&:=K_3\Big(\tfrac{12\theta^2}{\eta_1}+\tfrac{12\theta^2\widetilde{K}_\phi^2}{\eta_2}\Big)+\tfrac{A_1}{4\varepsilon}+\tfrac{32\theta^2}{\eta_1},\\
K_7&:=3K_3A_2+\tfrac{3(\varepsilon+3A_2)}{4},\\
K_8&:=K_3\tfrac{3(K_f+A_1)^2}{\varepsilon^2A_2}+\tfrac{3(K_f+A_1)^2}{2\varepsilon^2A_2},\\
K_9&:=K_3\Big(\tfrac{12\theta^2\widetilde{K}_\phi^2}{\eta_1}+\tfrac{3(K_h+B)^2}{\eta_2}\Big)+\tfrac{32\theta^2\widetilde{K}_\phi^2}{\eta_1},\\
K_{10}&:=K_3\Big(\tfrac{12\theta^2}{\eta_1}+\tfrac{12\theta^2\widetilde{K}_\phi^2}{\eta_2}\Big)+\tfrac{32\theta^2}{\eta_1},\\
K_{11}&:=3K_3A_2+\tfrac{3A_2}{2}.
\end{aligned}
$$

Thus, combining (25) with (31), we get

$$
I_4+I_5\le K_4\tau\|e_\phi^n\|^2+K_5\tau\|e_\rho^n\|^2+K_6\tau\|\nabla_{\mathcal{N}}e_\phi^n\|^2
$$

$$+ K_7\tau\|\Delta_{\mathcal{N}} e_\phi^n\|^2 + \frac{\eta_1}{3}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_\phi^n\|^2 + K_{12}\tau\|e_{\phi_1}\|^2 + K_{13}\tau\|e_{\rho_1}\|^2 + K_{14}\tau\|\nabla_{\mathcal{N}} e_{\phi_1}\|^2 + K_{15}\tau\|\Delta_{\mathcal{N}} e_{\phi_1}\|^2, \tag{32}$$

where $K_{12} := \frac{3(K_f+A_1)^2}{2\varepsilon^2 A_2} + K_8$, $K_{13} := K_1 + K_9$, $K_{14} := K_2 + K_{10}$ and $K_{15} := \frac{3A_2}{2} + K_{11}$. Similarly for the estimate (27), we can obtain

$$\begin{aligned} &K_{12}\tau\|e_{\phi_1}\|^2 + K_{13}\tau\|e_{\rho_1}\|^2 + K_{14}\tau\|\nabla_{\mathcal{N}} e_{\phi_1}\|^2 + K_{15}\tau\|\Delta_{\mathcal{N}} e_{\phi_1}\|^2 \\ &\le K_{16}\tau\|e_{\phi_1}\|^2 + K_{16}\tau\|e_{\rho_1}\|^2 + \frac{2\varepsilon}{3}\tau\|\Delta_{\mathcal{N}} e_{\phi_1}\|^2 + \frac{\eta_1}{8}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_{\phi_1}\|^2, \end{aligned} \tag{33}$$

where $K_{16} := \max\left\{K_{12} + \frac{3}{8\varepsilon}\left(K_{14} + \frac{2K_{15}^2}{\eta_1}\right)^2, K_{13}\right\}$.

Part 3. To obtain an estimate for the right-hand side of (33), we turn to the error equations (8a) and (8b), which yields (36), the estimate for $I_3 + I_4 + I_5$.

Acting $(I - \tau\mathcal{L}_1)$ and $(I - \tau\mathcal{L}_2)$ on both sides of (8a) and (8b) respectively, taking the discrete L^2 inner product on the first derived equation with e_{ϕ_1}, the second derived equation with e_{ϕ_1}, and deducing similarly as those done for (8e) and (8f) result in

$$\begin{aligned} \|e_{\phi_1}\|^2 + \|e_{\rho_1}\|^2 &\le \|e_\phi^n\|^2 + \|e_\rho^n\|^2 + \frac{4(K_f+A_1)^2}{\varepsilon^2 A_2}\tau\|e_\phi^n\|^2 + 4A_2\tau\|\Delta_{\mathcal{N}} e_\phi^n\|^2 \\ &+ \Big(\frac{16\theta^2\widetilde{K}_\phi^2}{\eta_1} + \frac{4(K_h+B)^2}{\eta_2}\Big)\tau\|e_\rho^n\|^2 + \Big(\frac{16\theta^2}{\eta_1} + \frac{16\theta^2\widetilde{K}_\phi^2}{\eta_2}\Big)\tau\|\nabla_{\mathcal{N}} e_\phi^n\|^2. \end{aligned} \tag{34}$$

Acting $-\tau\mathcal{L}_1$ on both sides of (8a) and taking the discrete L^2 inner product on the resulted equation with e_{ϕ_1}, and deriving similarly as those done for (8c), we have

$$\begin{aligned} &\frac{2\varepsilon}{3}\tau\|\Delta_{\mathcal{N}} e_{\phi_1}\|^2 + \frac{\eta_1}{8}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_{\phi_1}\|^2 \\ &\le \frac{A_1}{4\varepsilon}\tau\|\nabla_{\mathcal{N}} e_\phi^n\|^2 + \frac{3(\varepsilon+A_2)}{4}\tau\|\Delta_{\mathcal{N}} e_\phi^n\|^2 + \frac{\eta_1}{3}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_\phi^n\|^2 \\ &+ \frac{3(K_f+A_1)^2}{4\varepsilon^2 A_2}\tau\|e_\phi^n\|^2 + \frac{16\theta^2\widetilde{K}_\phi^2}{\eta_1}\tau\|e_\rho^n\|^2 + \frac{16\theta^2}{\eta_1}\tau\|\nabla_{\mathcal{N}} e_\phi^n\|^2 \\ &+ \frac{3A_2}{4}\tau\|\Delta_{\mathcal{N}} e_\phi^n\|^2. \end{aligned} \tag{35}$$

Combining (24), (32), (33), (34) and (35), if $\tau < 1$, we have

$$\begin{aligned} I_3 + I_4 + I_5 &\le K_{17}\tau\|e_\phi^n\|^2 + K_{18}\tau\|e_\rho^n\|^2 + K_{19}\tau\|\nabla_{\mathcal{N}} e_\phi^n\|^2 \\ &+ K_{20}\tau\|\Delta_{\mathcal{N}} e_\phi^n\|^2 + \frac{2\eta_1}{3}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}} e_\phi^n\|^2, \end{aligned} \tag{36}$$

where

$$\begin{aligned} K_{17} &:= \tfrac{3(K_f+A_1)^2}{2\varepsilon^2 A_2} + K_4 + K_{16} + K_{16}\tfrac{4(K_f+A_1)^2}{\varepsilon^2 A_2} + \tfrac{3(K_f+A_1)^2}{4\varepsilon^2 A_2}, \\ K_{18} &:= K_1 + K_5 + K_{16} + K_{16}\Big(\tfrac{16\theta^2\widetilde{K}_\phi^2}{\eta_1} + \tfrac{4(K_h+B)^2}{\eta_2}\Big) + \tfrac{16\theta^2\widetilde{K}_\phi^2}{\eta_1}, \\ K_{19} &:= K_2 + K_6 + K_{16}\Big(\tfrac{16\theta^2}{\eta_1} + \tfrac{16\theta^2\widetilde{K}_\phi^2}{\eta_2}\Big) + \tfrac{A_1}{4\varepsilon} + \tfrac{16\theta^2}{\eta_1}, \\ K_{20} &:= K_7 + 4K_{16}A_2 + \tfrac{3(\varepsilon+4A_2)}{4}. \end{aligned}$$

Note that

$$K_{19}\tau\|\nabla_{\mathcal{N}}e_{\phi}^{n}\|^{2}+K_{20}\tau\|\Delta_{\mathcal{N}}e_{\phi}^{n}\|^{2}$$
$$\leq\frac{1}{4\varepsilon}\Big(K_{19}+\frac{3K_{20}^{2}}{\eta_{1}}\Big)^{2}\tau\|e_{\phi}^{n}\|^{2}+\varepsilon\tau\|\Delta_{\mathcal{N}}e_{\phi}^{n}\|^{2}+\frac{\eta_{1}}{12}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi}^{n}\|^{2}. \tag{37}$$

Part 4. We combine all the estimates obtained in the previous parts to derive the result of error estimate.

Plugging (36) and (37) into (23), we get

$$\frac{1}{2}\|e_{\phi}^{n+1}\|^{2}+\frac{1}{2}\|e_{\rho}^{n+1}\|^{2}+\frac{A_{1}}{\varepsilon}\tau\|\nabla_{\mathcal{N}}e_{\phi}^{n+1}\|^{2}+B\tau\|\nabla_{\mathcal{N}}e_{\rho}^{n+1}\|^{2}$$
$$+\varepsilon\tau\|\Delta_{\mathcal{N}}e_{\phi}^{n+1}\|^{2}+\frac{3\eta_{1}}{4}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi}^{n+1}\|^{2}$$
$$\leq\frac{1}{2}\|e_{\phi}^{n}\|^{2}+\frac{1}{2}\|e_{\rho}^{n}\|^{2}+\frac{\tau}{4}\|e_{\phi}^{n+1}\|^{2}+\frac{\tau}{4}\|e_{\rho}^{n+1}\|^{2}+(C_{1}^{2}+C_{2}^{2})\tau^{7}$$
$$+K_{21}\tau\|e_{\phi}^{n}\|^{2}+K_{21}\tau\|e_{\rho}^{n}\|^{2}+\varepsilon\tau\|\Delta_{\mathcal{N}}e_{\phi}^{n}\|^{2}+\frac{3\eta_{1}}{4}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi}^{n}\|^{2},$$

where $K_{21}:=\max\big\{K_{17}+\frac{1}{4\varepsilon}\big(K_{19}+\frac{3K_{20}^{2}}{\eta_{1}}\big)^{2},K_{18}\big\}$. Defining

$$F^{n+1}=\frac{1}{2}\|e_{\phi}^{n+1}\|^{2}+\frac{1}{2}\|e_{\rho}^{n+1}\|^{2}+\frac{A_{1}}{\varepsilon}\tau\|\nabla_{\mathcal{N}}e_{\phi}^{n+1}\|^{2}+\varepsilon\tau\|\Delta_{\mathcal{N}}e_{\phi}^{n+1}\|^{2}$$
$$+\frac{3\eta_{1}}{4}\tau\|\nabla_{\mathcal{N}}\Delta_{\mathcal{N}}e_{\phi}^{n+1}\|^{2},$$

we have $F^{n+1}-F^{n}\leq\frac{1}{2}\tau F^{n+1}+2K_{21}\tau F^{n}+(C_{1}^{2}+C_{2}^{2})\tau^{7}$. If $\tau<1$, summing the above inequality from 0 to n and applying the discrete Gronwall's inequality to the resulted inequality, we obtain

$$F^{n+1}\leq 2(C_{1}^{2}+C_{2}^{2})Te^{(1+4K_{21})T}\tau^{6}. \tag{38}$$

Thus, we have

$$\|e_{\phi}^{n+1}\|+\|e_{\rho}^{n+1}\|\leq C\tau^{3}, \tag{39}$$

where C is a positive constant depending on C_1, C_2, K_{21} and T, which finishes the error estimate.

Next we prove the bound-preserving property (11). First, for ϕ^{n+1} and ρ^{n+1}, applying the discrete inverse inequality to (39), we obtain

$$\|e_{\phi}^{n+1}\|_{\infty}+\|e_{\rho}^{n+1}\|_{\infty}\leq C_{i}h_{x}^{-\frac{1}{2}}h_{y}^{-\frac{1}{2}}(\|e_{\phi}^{n+1}\|+\|e_{\rho}^{n+1}\|)\leq C_{i}Ch_{x}^{-\frac{1}{2}}h_{y}^{-\frac{1}{2}}\tau^{3},$$

where C_i is the constant derived by the inverse inequality. Thus, if $h_xh_y\leq\frac{\delta_0}{2C_iCC_0^3}$ with the application of the linear refinement path constraint $\tau\leq C_0h_x^{\frac{1}{2}}h_y^{\frac{1}{2}}$, we have

$$\|e_{\phi}^{n+1}\|_{\infty}+\|e_{\rho}^{n+1}\|_{\infty}\leq C_{i}CC_{0}^{3}h_{x}h_{y}\leq\frac{\delta_{0}}{2}. \tag{40}$$

Since the numerical solution can be regarded as the sum of the exact solution and the numerical error, a combination of (9) and (40) leads to the bound-preserving property for ϕ^n and ρ^n in (11). For the boundedness of $|\nabla_{\mathcal{N}}\phi^{n+1}|$, from (38) we can obtain $\|\nabla_{\mathcal{N}}e_{\phi}^{n+1}\|\leq C'\tau^{\frac{5}{2}}$,

where C' is a constant depending on $A_1, \varepsilon, C_1, C_2, K_{21}$ and T. Applying the discrete inverse inequality yields

$$\|\nabla_{\mathcal{N}} e_\phi^{n+1}\|_\infty \le C_i' h_x^{-\frac{1}{2}} h_y^{-\frac{1}{2}} \|\nabla_{\mathcal{N}} e_\phi^{n+1}\| \le C_i' C' h_x^{-\frac{1}{2}} h_y^{-\frac{1}{2}} \tau^{\frac{5}{2}}.$$

If $h_x h_y \le \left(C_i' C' C_0^{\frac{5}{2}}\right)^{-\frac{4}{3}}$ and $\tau \le C_0 h_x^{\frac{1}{2}} h_y^{\frac{1}{2}}$, we have

$$\|\nabla_{\mathcal{N}} e_\phi^{n+1}\|_\infty \le C_i' C' C_0^{\frac{5}{2}} h_x^{\frac{3}{4}} h_y^{\frac{3}{4}} \le 1. \tag{41}$$

Combining (9) with (41), we obtain the boundedness for $|\nabla_{\mathcal{N}} \phi^{n+1}|$ in (11). By completing the induction, we finish the proof. □

Remark 5 Since $\varphi_1(\tau\mathcal{L}_k)$ and $\widetilde{\varphi}_l(\tau\mathcal{L}_k)$ ($l = 2, 3, 4, 5, 6$ and $k = 1, 2$) are well-defined only on $\mathcal{M}_{\mathcal{N}}^0$, additional steps are required when applying the integration-by-parts formula to the inner product containing nonlinear or coupling terms which do not belong to $\mathcal{M}_{\mathcal{N}}^0$ in the convergence analysis. For instance, denoting $\psi = f'(\phi^{t_n}) - f'(\phi^n)$ and $\bar{\psi} = \frac{1}{N_x N_y}\langle \psi, 1\rangle$, we have

$$\begin{aligned} I_{131} &= \frac{\tau}{\varepsilon}\left\langle (I - \tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\Delta_{\mathcal{N}}\psi, e_\phi^{n+1}\right\rangle \\ &= \frac{\tau}{\varepsilon}\left\langle (I - \tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\Delta_{\mathcal{N}}\left(\psi - \bar{\psi}\right), e_\phi^{n+1}\right\rangle \\ &= \frac{\tau}{\varepsilon}\left\langle (I - \tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\left(\psi - \bar{\psi}\right), \Delta_{\mathcal{N}} e_\phi^{n+1}\right\rangle \\ &\le \frac{\tau}{\varepsilon}\|\psi\|_{\mathcal{N}}\|\Delta_{\mathcal{N}} e_\phi^{n+1}\|_{\mathcal{N}}, \end{aligned}$$

where we have used the fact that $\|\psi - \bar{\psi}\| \le \|\psi\|$, which is derived from the Parseval's identity. Since $\langle \psi - \bar{\psi}, 1\rangle = 0$, we know $(I - \tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)$ is well-defined such that $(I - \tau\mathcal{L}_1)\widetilde{\varphi}_4(\tau\mathcal{L}_1)\left(\psi - \bar{\psi}\right)$ makes sense. These additional steps guarantee the validity of the multiplication involving $\varphi_1, \widetilde{\varphi}_l$ ($l = 2, 3, 4, 5, 6$) and terms outside of $\mathcal{M}_{\mathcal{N}}^0$, which have been omitted in the proof for simplicity.

Denote by $(\phi_e(\boldsymbol{x}, t), \rho_e(\boldsymbol{x}, t))$ the exact solution of the system (2). To obtain the error estimate between the space-discrete solution $(\phi(t), \rho(t))$ and the exact solution, we define $(\Phi(t), \Psi(t))$ the restriction of $(\phi_e(\boldsymbol{x}, t), \rho_e(\boldsymbol{x}, t))$ on the mesh $\mathcal{M}_{\mathcal{N}}$. Define $H_{\text{per}}^m(\Omega) = \{u \in H^m(\Omega) \,|\, u \text{ is } \Omega\text{-periodic}\}$. By performing the similar analysis as in [8, 22], we have the following error estimate.

Theorem 2 *Suppose $\phi_e \in L^\infty(0, T; H_{per}^{m+6}(\Omega))$ and $\rho_e \in L^\infty(0, T; H_{per}^{m+4}(\Omega))$. For any fixed $t \in [0, T]$, we have*

$$\|\Phi(t) - \phi(t)\| + \|\Psi(t) - \rho(t)\| \le C^* h^m, \quad t \in [0, T],$$

where $h = \max\{h_x, h_y\}$ and C^* is a positive constant independent of h.

Based on the Theorem 2, the regularity assumption $\phi_e \in H^3(0, T; H_{\text{per}}^{10}(\Omega))$ and $\rho_e \in H^3(0, T; H_{\text{per}}^9(\Omega))$ imply $\Delta_{\mathcal{N}}^5 \phi \in H^3(0, T; \mathcal{M}_{\mathcal{N}})$ and $\nabla_{\mathcal{N}} \Delta_{\mathcal{N}}^4 \rho \in H^3(0, T; \mathcal{M}_{\mathcal{N}})$. Therefore, combining Theorems 1 and 2, we obtain the following theorem on the error estimate of the fully-discrete ETDRK3 scheme.

Theorem 3 *Suppose that*

$$\begin{cases} \phi_e \in H^3(0,T;H^{10}_{per}(\Omega)) \cap L^\infty(0,T;H^{m+6}_{per}(\Omega)), \\ \rho_e \in H^3(0,T;H^{9}_{per}(\Omega)) \cap L^\infty(0,T;H^{m+4}_{per}(\Omega)), \end{cases}$$

with $m \geq 2$ and $\{(\phi^n, \rho^n), 0 \leq n \leq N_t\}$ is the numerical solution derived by the ETDRK3 scheme (6). If h_x, h_y and τ are sufficiently small subject to a constraint $\tau \leq \min\{1, C_0 h_x^{\frac{1}{2}} h_y^{\frac{1}{2}}\}$, we have

$$\|\Phi(t_n) - \phi^n\| + \|\Psi(t_n) - \rho^n\| \leq \widetilde{C}(\tau^3 + h^m), \quad 0 \leq n \leq N_t,$$

where $\widetilde{C} > 0$ is a constant independent of h and τ, and the numerical solution ϕ^n and ρ^n satisfy the the bound-preserving property (11).

5 Discrete Energy Dissipation

Based on the linear convex splitting of the discrete energy (4), we obtain the following result for the discrete energy of the ETDRK3 scheme (6).

Theorem 4 (Discrete energy dissipation for ETDRK3) *Under the conditions in Theorem 3, if A_1, A_2, and B satisfy (3), then the numerical scheme (6) is energy dissipative, i.e.,*

$$E(\phi^{n+1}, \rho^{n+1}) \leq E(\phi^n, \rho^n), \qquad 0 \leq n \leq N_t. \tag{42}$$

Proof Under the conditions in Theorem 3, if A_1, A_2, and B satisfy (3), we know $e_e(\boldsymbol{v})$ is convex according to Proposition 3. Denote $\phi_0 = \phi^n$, $\rho_0 = \rho^n$, $\phi_3 = \phi^{n+1}$ and $\rho_3 = \rho^{n+1}$, and recall ϕ_1, ρ_1, ϕ_2 and ρ_2 given by (6a)-(6d). Following the similar spirit from [35], we have the inequality

$$\begin{aligned} -E_e(\phi_{k+1}, \rho_{k+1}) + E_e(\phi_k, \rho_k) \leq\ & \langle \Delta_{\mathcal{N}}^{-1} F_1(\phi_k, \rho_k), \phi_{k+1} - \phi_k \rangle \\ & + \langle \Delta_{\mathcal{N}}^{-1} F_2(\phi_k, \rho_k), \rho_{k+1} - \rho_k \rangle . \end{aligned} \tag{43}$$

For E_c, we have

$$\begin{aligned} E_c(\phi_{k+1}, \rho_{k+1}) - E_c(\phi_k, \rho_k) &= \frac{1}{2}\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_1\phi_{k+1}, \phi_{k+1}\rangle + \frac{1}{2}\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_2\rho_{k+1}, \rho_{k+1}\rangle \\ &\quad - \frac{1}{2}\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_1\phi_k, \phi_k\rangle - \frac{1}{2}\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_2\rho_k, \rho_k\rangle \\ &= \big\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_1\phi_{k+1}, \phi_{k+1} - \phi_k\big\rangle - \frac{1}{2}\big\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_1(\phi_{k+1} - \phi_k), \phi_{k+1} - \phi_k\big\rangle \\ &\quad + \big\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_2\rho_{k+1}, \rho_{k+1} - \rho_k\big\rangle - \frac{1}{2}\big\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_2(\rho_{k+1} - \rho_k), \rho_{k+1} - \rho_k\big\rangle . \end{aligned} \tag{44}$$

Since $E(\phi_k, \rho_k) = E_c(\phi_k, \rho_k) - E_e(\phi_k, \rho_k)$, by combing with (43) and (44) we obtain

$$\begin{aligned} &E(\phi^{n+1}, \rho^{n+1}) - E(\phi^n, \rho^n) = \textstyle\sum_{k=0}^{2} \big[E(\phi_{k+1}, \rho_{k+1}) - E(\phi_k, \rho_k)\big] \\ &\leq \textstyle\sum_{k=0}^{2} \big[\big\langle \Delta_{\mathcal{N}}^{-1}(\mathcal{L}_1\phi_{k+1} + F_1(\phi_k, \rho_k) - \frac{1}{2}\mathcal{L}_1(\phi_{k+1} - \phi_k)), \phi_{k+1} - \phi_k\big\rangle \\ &\qquad + \big\langle \Delta_{\mathcal{N}}^{-1}(\mathcal{L}_2\rho_{k+1} + F_2(\phi_k, \rho_k) - \frac{1}{2}\mathcal{L}_2(\rho_{k+1} - \rho_k)), \rho_{k+1} - \rho_k\big\rangle\big]. \end{aligned} \tag{45}$$

Note that the ETDRK3 scheme (6) can be written as the following form:

$$\phi_i = \phi_0 + \tau \textstyle\sum_{j=0}^{i-1} a_{ij}(\tau\mathcal{L}_1)(\mathcal{L}_1\phi_0 + F_1(\phi_j, \rho_j)), \quad i = 1, 2,$$

$$\rho_i = \rho_0 + \tau \sum_{j=0}^{i-1} a_{ij}(\tau\mathcal{L}_2)(\mathcal{L}_2\rho_0 + F_2(\phi_j, \rho_j)), \quad i = 1, 2,$$
$$\phi_3 = \phi_0 + \tau \sum_{j=0}^{2} b_j(\tau\mathcal{L}_1)(\mathcal{L}_1\phi_0 + F_1(\phi_j, \rho_j)),$$
$$\rho_3 = \rho_0 + \tau \sum_{j=0}^{2} b_j(\tau\mathcal{L}_2)(\mathcal{L}_2\phi_0 + F_2(\phi_j, \rho_j)),$$

where $a_{10}, a_{20}, a_{21}, b_0, b_1$ and b_2 can be constructed similarly as in [20]. Denoting

$$P = \begin{pmatrix} a_{10} & 0 & 0 \\ a_{20} & a_{21} & 0 \\ b_0 & b_1 & b_2 \end{pmatrix}, \quad E_L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix},$$

we have

$$\begin{pmatrix} \phi_1 - \phi_0 \\ \phi_2 - \phi_0 \\ \phi_3 - \phi_0 \end{pmatrix} = \tau P \begin{pmatrix} \mathcal{L}_1\phi_0 + F_1(\phi_0, \rho_0) \\ \mathcal{L}_1\phi_0 + F_1(\phi_1, \rho_1) \\ \mathcal{L}_1\phi_0 + F_1(\phi_2, \rho_2) \end{pmatrix}.$$

Therefore,

$$\begin{pmatrix} \mathcal{L}_1\phi_0 + F_1(\phi_0, \rho_0) \\ \mathcal{L}_1\phi_0 + F_1(\phi_1, \rho_1) \\ \mathcal{L}_1\phi_0 + F_1(\phi_2, \rho_2) \end{pmatrix} = \frac{1}{\tau} P^{-1} \begin{pmatrix} \phi_1 - \phi_0 \\ \phi_2 - \phi_0 \\ \phi_3 - \phi_0 \end{pmatrix} = \frac{1}{\tau} P^{-1} E_L \begin{pmatrix} \phi_1 - \phi_0 \\ \phi_2 - \phi_1 \\ \phi_3 - \phi_2 \end{pmatrix}.$$

Thus, we obtain

$$\begin{aligned}
&\sum_{k=0}^{2} \left\langle \Delta_{\mathcal{N}}^{-1}(\mathcal{L}_1\phi_{k+1} + F_1(\phi_k, \rho_k) - \tfrac{1}{2}\mathcal{L}_1(\phi_{k+1} - \phi_k)), \phi_{k+1} - \phi_k \right\rangle \\
&= \sum_{k=0}^{2} \left\langle \Delta_{\mathcal{N}}^{-1}(\mathcal{L}_1\phi_{k+1} - \mathcal{L}_1\phi_0 + \mathcal{L}_1\phi_0 + F_1(\phi_k, \rho_k)), \phi_{k+1} - \phi_k \right\rangle \\
&\quad - \frac{1}{2} \sum_{k=0}^{2} \left\langle \Delta_{\mathcal{N}}^{-1}\mathcal{L}_1(\phi_{k+1} - \phi_k), \phi_{k+1} - \phi_k \right\rangle \\
&= \frac{1}{\tau} \sum_{k=0}^{2} \sum_{j=0}^{k} \Big[\left\langle \Delta_{\mathcal{N}}^{-1}\tau\mathcal{L}_1(\phi_{j+1} - \phi_j), \phi_{k+1} - \phi_k \right\rangle \\
&\qquad + \left\langle \Delta_{\mathcal{N}}^{-1}(P^{-1}E_L)_{kj}(\phi_{j+1} - \phi_j), \phi_{k+1} - \phi_k \right\rangle \Big] \\
&\quad - \frac{1}{2\tau} \sum_{k=0}^{2} \left\langle \Delta_{\mathcal{N}}^{-1}\tau\mathcal{L}_1(\phi_{k+1} - \phi_k), \phi_{k+1} - \phi_k \right\rangle \\
&= \frac{1}{\tau} \sum_{k,j=0}^{2} \left\langle \Delta_{\mathcal{N}}^{-1} D_{kj}(\tau\mathcal{L}_1)(\phi_{j+1} - \phi_j), \phi_{k+1} - \phi_k \right\rangle,
\end{aligned} \tag{46}$$

where $D(z) = zE_L + P^{-1}E_L - \frac{z}{2}I$ with $z = \tau\mathcal{L}_1$. Similarly,

$$\begin{aligned}
&\sum_{k=0}^{2} \left\langle \Delta_{\mathcal{N}}^{-1}(\mathcal{L}_2\rho_{k+1} + F_2(\phi_k, \rho_k) - \tfrac{1}{2}\mathcal{L}_2(\rho_{k+1} - \rho_k)), \rho_{k+1} - \rho_k \right\rangle \\
&= \frac{1}{\tau} \sum_{k,j=0}^{2} \left\langle \Delta_{\mathcal{N}}^{-1} D_{kj}(\tau\mathcal{L}_2)(\rho_{j+1} - \rho_j), \rho_{k+1} - \rho_k \right\rangle.
\end{aligned} \tag{47}$$

A combination of (45), (46), and (47) leads to

$$\begin{aligned}
&E(\phi^{n+1}, \rho^{n+1}) - E(\phi^n, \rho^n) \\
&\le \frac{1}{\tau} \sum_{k,j=0}^{2} \Big(\left\langle \Delta_{\mathcal{N}}^{-1} D_{kj}(\tau\mathcal{L}_1)(\phi_{j+1} - \phi_j), \phi_{k+1} - \phi_k \right\rangle \\
&\qquad + \left\langle \Delta_{\mathcal{N}}^{-1} D_{kj}(\tau\mathcal{L}_2)(\rho_{j+1} - \rho_j), \rho_{k+1} - \rho_k \right\rangle \Big).
\end{aligned}$$

It has been proved in [15] that the $D(z)$ associated with the scheme (6) is positive-definite for all negative $z \in \mathbb{R}$. According to Proposition 4, we know that $\phi_{j+1} - \phi_j$ and $\rho_{j+1} - \rho_j$ belong to $\mathcal{M}_{\mathcal{N}}^0$ for $j = 0, 1, 2$. Since $\mathcal{L}_1$ and $\mathcal{L}_2$ are negative definite on $\mathcal{M}_{\mathcal{N}}^0$, it follows

from Lemma 1 that both $D(\tau\mathcal{L}_1)$ and $D(\tau\mathcal{L}_2)$ are positive definite. Hence, the energy stable property is obtained by the negative definiteness of $\Delta_{\mathcal{N}}^{-1}$. □

6 Numerical Experiments

In this section, we perform some numerical experiments to verify the accuracy and structure-preserving property of the proposed ETDRK3 scheme (6), and simulate the phase separation phenomena of the BFS system. By considering the periodic boundary condition, the spectral collocation approximations for spatial discretization can be implemented using the fast Fourier transform as in [8].

To ensure the energy stability of the scheme, we choose the stabilization parameters based on the conditions in (3). The parameters that are smaller than the lower bounds in (3) might lead to an unstable scheme. Since the bounds of ϕ^n, $|\nabla_{\mathcal{N}}\phi^n|$ and ρ^n are difficult to be determined in advance, we usually determine their approximated bounds based on the experience and some numerical tests, and then take the values of the stabilization parameters according to (3). It should be noted that larger stabilization parameters can lead to larger numerical errors, which will be shown in the following convergence tests.

6.1 Convergence Tests

We consider the domain $\Omega = (0, 2\pi)^2$ and the following initial data

$$\begin{cases} \phi_0(x, y) = 0.2\sin(2x)\sin(2y), \\ \rho_0(x, y) = 0.5 + 0.1\cos(2x)\cos(2y). \end{cases}$$

The parameters are chosen as

$$\begin{aligned} &\eta_1 = 0.01, \quad \eta_2 = 0.05, \quad \varepsilon = 0.1, \quad \theta = 0.0005, \quad \beta = 0.02, \\ &\beta_1 = 0.03, \quad A_1 = 2.5, \quad A_2 = 0.01, \quad B = 0.2, \end{aligned}$$

and we set the 256×256 spatial mesh. Since the exact solution of the system (2) is unknown, the solution of the ETDRK3 scheme (6) with time step size $\tau = 10^{-6}$ is adopted as the benchmark solution. Then, we calculate the numerical solutions of the scheme (6) with different time step sizes $\tau = 0.01 \times 2^{-k}$ ($k = 7, 8, \ldots, 11$) and the differences between the benchmark and numerical solutions which are regarded as numerical errors. The discrete L^2 norms of the numerical errors at $t = 0.1$ are shown in Table 1 which verify the proposed third-order accuracy (in time) of the ETDRK3 scheme (6). Moreover, we perform the convergence test by choosing larger stabilization parameters, i.e., $A_1 = 12.5$, $A_2 = 0.05$, $B = 1$, with other parameters unchanged, and present the numerical errors in Table 2. It is shown that the larger stabilization parameters result in larger numerical errors. Thus, to maintain both the stability and accuracy of the scheme, we should avoid taking excessively large parameter values, provided they meet the conditions in (3).

Table 1 Temporal L^2-errors and convergence rates of ϕ and ρ with $A_1 = 2.5$, $A_2 = 0.01$, $B = 0.2$.

τ	Error(ϕ)	Rate(ϕ)	Error(ρ)	Rate(ρ)
0.01×2^{-7}	1.15×10^{-8}	–	9.08×10^{-9}	–
0.01×2^{-8}	1.71×10^{-9}	2.75	1.30×10^{-9}	2.81
0.01×2^{-9}	2.41×10^{-10}	2.82	1.76×10^{-10}	2.88
0.01×2^{-10}	3.26×10^{-11}	2.89	2.32×10^{-11}	2.92
0.01×2^{-11}	4.13×10^{-12}	2.98	2.98×10^{-12}	2.97

Table 2 Temporal L^2-errors and convergence rates of ϕ and ρ with $A_1 = 12.5$, $A_2 = 0.05$, $B = 1$.

τ	Error(ϕ)	Rate(ϕ)	Error(ρ)	Rate(ρ)
0.01×2^{-7}	4.49×10^{-7}	–	2.98×10^{-7}	–
0.01×2^{-8}	6.71×10^{-8}	2.74	4.50×10^{-8}	2.73
0.01×2^{-9}	9.37×10^{-9}	2.84	6.28×10^{-9}	2.84
0.01×2^{-10}	1.25×10^{-9}	2.91	8.34×10^{-10}	2.91
0.01×2^{-11}	1.61×10^{-10}	2.95	1.07×10^{-10}	2.96

6.2 The Effect of Stabilization Parameters on Simulations

We consider $\Omega = (0, 2\pi)^2$, $N_x = N_y = 256$, $\tau = 0.001$, and the following initial data:

$$\begin{aligned}\phi_0(x, y) &= \tanh\left(\frac{0.4\pi - \sqrt{(x - 0.7\pi)^2 + (y - \pi)^2}}{\sqrt{2}\varepsilon}\right) \\ &\quad + \tanh\left(\frac{0.17\pi - \sqrt{(x - 1.4\pi)^2 + (y - \pi)^2}}{\sqrt{2}\varepsilon}\right) + 1, \\ \rho_0(x, y) &= 0.02 + 0.2(1 - |\phi_0(x, y)|).\end{aligned}$$

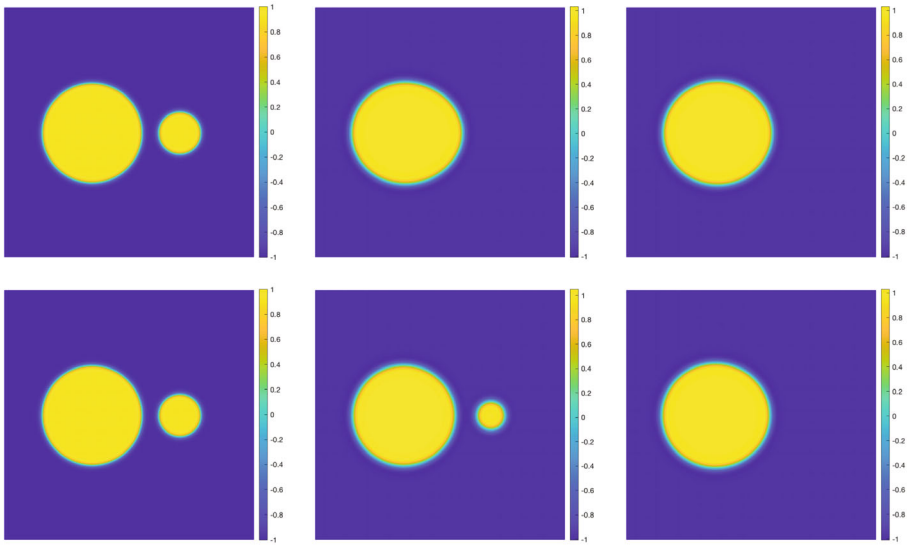
The parameters are chosen as

$$\eta_1 = 0.0002, \quad \eta_2 = 0.0004, \quad \varepsilon = 0.04, \quad \theta = 0.0015, \quad \beta = 0.05, \quad \beta_1 = 0.15.$$

We plot the snapshots of droplets merging taken at different times with respect to $A_1 = 4$, $A_2 = 0.05$, $B = 6$ and $A_1 = 20$, $A_2 = 0.25$, $B = 30$ in Figure 1, and the evolutions of the energy in Figure 2. As shown in Figure 1, the two droplets merge more slowly when larger stabilization parameters are employed. This delaying effect is a consequence of the larger numerical errors generated, as observed in the convergence tests. It is also shown in Figure 2 that the energy decays slower with larger stabilization parameters adopted, which is consistent with the snapshots in Figure 1.

6.3 3D Phase Separation Simulations

We carry out the simulation of the phase separation. As a homogeneous mixture is quenched into an unstable state, the phase separation, which is also called spinodal decomposition, will happen and lead the whole system to the two-phase state. Here, we consider $\Omega = (0, 2\pi)^3$



(a) $t = 0$ (b) $t = 0.5$ (c) $t = 1$

Fig. 1 Snapshots of droplets merging taken at different times with different choices of the stabilization parameters. The first row corresponds to $A_1 = 4$, $A_2 = 0.05$, $B = 6$, and the second row corresponds to $A_1 = 20$, $A_2 = 0.25$, $B = 30$

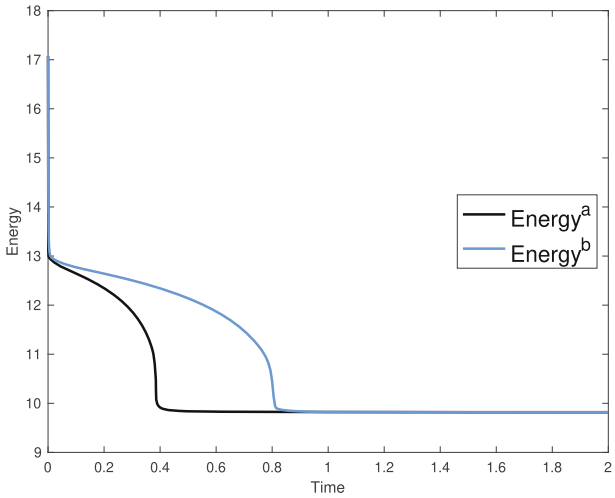
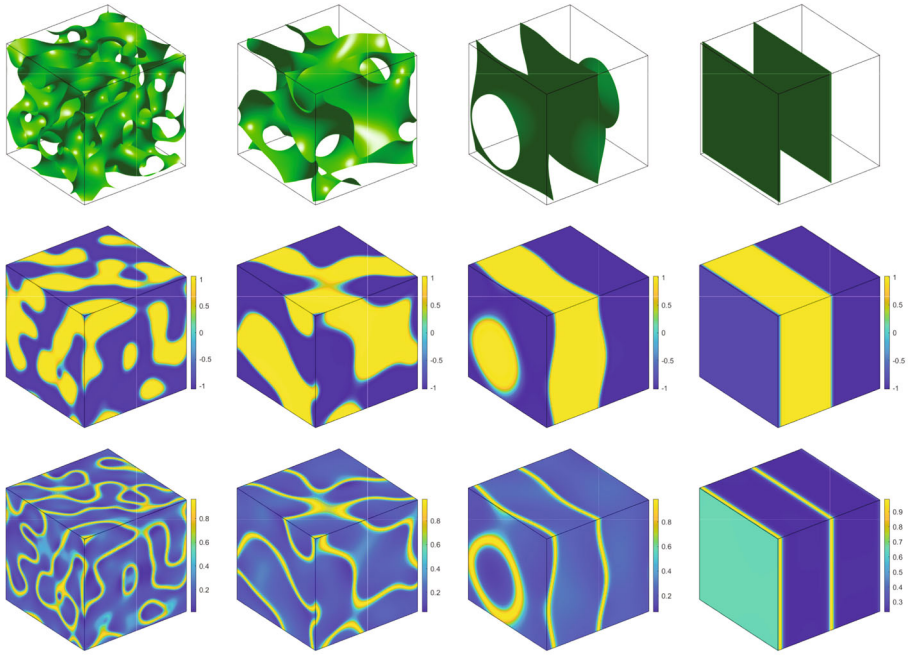


Fig. 2 Evolutions of the energy with different choices of the stabilization parameters. The superscripts a and b correspond to the evolutions associated with $A_1 = 4$, $A_2 = 0.05$, $B = 6$, and $A_1 = 20$, $A_2 = 0.25$, $B = 30$, respectively

and the following random initial data to model the homogeneous state

$$\begin{cases} \phi_0(x, y, z) = \bar{\phi}_0 + 0.001\text{rand}(x, y, z), \\ \rho_0(x, y, z) = 0.3 + 0.001\text{rand}(x, y, z), \end{cases}$$



(a) $t = 0.5$ (b) $t = 3$ (c) $t = 12$ (d) $t = 30$

Fig. 3 Snapshots of 3D phase separation simulation taken at different times for random initial data of $\bar{\phi}_0 = 0$. The first row shows the isosurface of $\phi = 0$, and the second and third rows show the snapshots of ϕ^n and ρ^n respectively

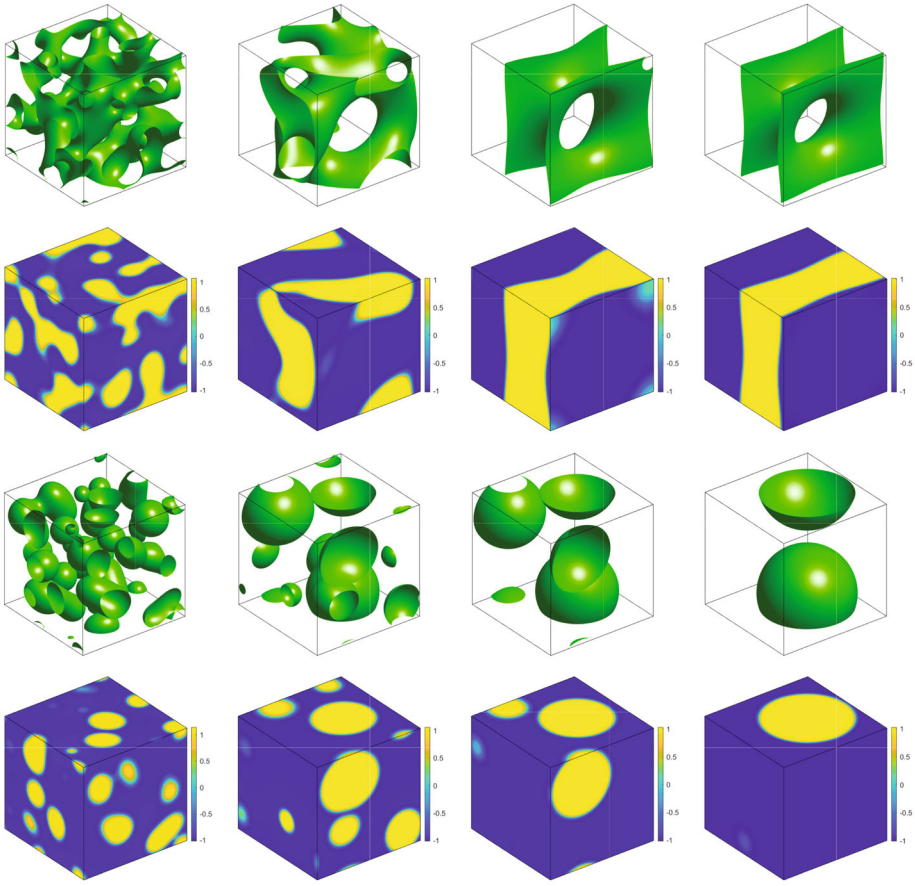
where rand(x, y, z) represents a random number uniformly distributed in $[-1, 1]$. The $128 \times 128 \times 128$ spatial mesh, time step size $\tau = 0.001$, and the parameters

$$\eta_1 = 0.0002, \quad \eta_2 = 0.0004, \quad \varepsilon = 0.04, \quad \theta = 0.0015,$$
$$\beta = 0.05, \quad \beta_1 = 0.15, \quad A_1 = 4, \quad A_2 = 0.05, \quad B = 6,$$

are taken. The computations are conducted in MATLAB on a computer with an Intel 2.90GHz Processor and 32GB Memory.

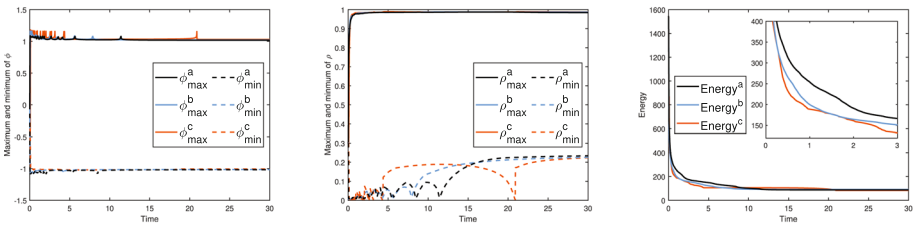
First, we choose $\bar{\phi}_0 = 0$. The snapshots of ϕ^n and ρ^n at different times are plotted in Figure 3. The simulation up to $t = 30$ takes 14.32 hours on the computer mentioned above. It is shown in Figure 3 that the same component of the fluid accumulates while the different components separate from each other, and the surfactants move to the interfaces. The mixture changes from a homogeneous state to a two-phase state gradually. We also plot the isosurface of $\phi = 0$ in Figure 3. Due to the amphiphilic nature, the surfactants mainly concentrate at the fluid interfaces. Therefore, the figures of the isosurface of $\phi = 0$ also serve as the indicator of the surfactant distribution within the system. The setting $\bar{\phi}_0 = 0$ implies that two components of the fluid have the same mass, so the interface is formed as a plane to minimize the area of the interface, thus minimizing the interfacial energy.

Then, we perform the simulation with $\bar{\phi}_0 = -0.3$ and $\bar{\phi}_0 = -0.5$, and the snapshots of ϕ^n and the isosurface of $\phi = 0$ at different times are plotted in Figure 4. As the mass ratio of the two components becomes more imbalanced, the interface between the two components becomes more curved. The interface for $\bar{\phi}_0 = -0.5$ ultimately takes the form of a sphere,



(a) $t = 0.5$ (b) $t = 3$ (c) $t = 12$ (d) $t = 30$

Fig. 4 Snapshots of 3D phase separation simulation taken at different times for random initial data of $\bar{\phi}_0 = -0.3, -0.5$. The first two rows show the isosurface of $\phi = 0$ and the snapshots of ϕ^n for $\bar{\phi}_0 = -0.3$, while the last two rows depict the same for $\bar{\phi}_0 = -0.5$



(a) (b) (c)

Fig. 5 Evolutions of the maximum and minimum of ϕ^n and ρ^n, and the evolution of energy. The superscripts a, b and c correspond to the evolutions associated with $\bar{\phi}_0 = 0$, $\bar{\phi}_0 = -0.3$, and $\bar{\phi}_0 = -0.5$, respectively

which also minimizes the interfacial energy. The simulations with different values of $\bar{\phi}_0$ were also conducted in [37, 41], which demonstrate similar characteristics.

In Figure 5, we present the evolutions of the maximum and minimum of ϕ^n and ρ^n, and the evolution of energy for $\bar{\phi}_0 = 0, -0.3, -0.5$. It is shown that in three cases, ϕ^n is bounded, ρ^n is restricted in $(0, 1)$, and the energy exhibits a decreasing trend over time, which verify the structure-preserving property, i.e., bound-preserving property and energy stability of the constructed ETDRK3 scheme (6). Since the boundedness of $|\nabla_{\mathcal{N}}\phi^n|$ can be implied by the boundedness of ϕ^n, we omit the figure of its evolution for brevity. It should be noted that the black and blue solid lines in Figure 5(a), the dashed lines of three colors in Figure 5(a), and the solid lines of three colors in Figure 5(b) almost overlap. Figures 5(a) and 5(b) reveal oscillations of the maximum and minimum of ϕ and ρ, which actually correspond to changes of the phase domain and are therefore reasonable. For example, the ρ^c_{min} in Figure 5(b) approaches and then moves away from 0 after $t = 20$. This transition aligns with the disappearance of the droplet that exists at $t = 12$ but vanished by $t = 30$ as shown in the last two rows of Figure 4. From Figure 5(b), we know that the value of ρ^n could be quite close to 0 and 1. In this case, a large value needs to be taken for the stabilization parameter B by the condition in Proposition 3. To avoid large splitting errors, we used $B = 6$ which is smaller than the value satisfying the theoretical requirement. Nevertheless, it can be observed in Figure 5(c) that the energy of three examples still decreases.

7 Conclusions

In this paper, we propose a third-order structure-preserving ETDRK scheme for the binary fluid-surfactant phase field model. By introducing three stabilization terms, we conduct the linear convex splitting of the free energy functional to ensure the energy dissipation with respect to the original energy. In order to bound the nonlinear and coupling terms and to determine the appropriate values for the stabilization parameters, it is necessary to justify the bound-preserving property of the numerical solution. By treating the numerical solution as a combination of the exact solution and the numerical error, we conduct an inductive analysis of the convergence and bound-preserving property. The constructed scheme is linear and fully decoupled, and the fast Fourier transform can be applied for implementation in rectangular domains. The numerical experiments verify the accuracy and structure-preserving property of the proposed scheme and the simulations of 3D phase separation phenomena also demonstrate its efficiency.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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